EdgePlus: A Multiagent Reinforcement Learning Framework for Dynamic Task Allocation in 6G Edge Computing

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***Abstract*—In smart-city air-quality monitoring, millions of IoT sensors generate time-sensitive data that must be processed within strict end-to-end deadlines, under dynamic mobility conditions, and with strong security guarantees. In this study, we present *EdgePlus*, a unified sixth-generation (6G) framework that couples 1) *masked* cooperative PPO (CCPPO) enforcing feasibility at sampling time (routing order, slice quotas, resource limits), 2) a *constricted* particle swarm optimization (PSO) outer loop that escapes local optima under strict SLA gates, 3) a lightweight advisory layer (Nash sharing, Ziegler–Nichols-initialized PID, and leader election) that biases logits without breaking masks, and 4) a two-tier *blockchain-lite* ledger [edge proof-of-authority (PoA), cloud Byzantine fault-tolerant BFT)] whose reputation and debt signals feedback into state, reward, and masks. In a 100 km^2 city-scale simulator (ten clusters; three load regimes of 50 000, 100 000, and 150 000 tasks). Under light load, all methods reach 100% success with seconds-level latencies and no observable penalty from masking or security. At stress (150k), EdgePlus achieves the highest on-time completion rate (82.77%), highest throughput (206.93), lowest violation rate (17.23%), and lowest 15 percentile latency (L_{p50} of 53.23 s). Ablations confirmed that *feasibility masks dominate* performance: removing them increases violations by 5.5%–10.0% points and tail latency (L_{p95}) by 26–47 s, while advisory and PSO deliver smaller but consistent gains. The PID controller remains robust under $\pm$20% gain drift (settling $\approx$ 2.7–2.9 s) by design-time Jury certification. In security experiments, baseline confidentiality, integrity, and availability (CIA)/auth attacks are mitigated at 98%–99% and advanced (nonphysical-layer) threats achieve 70%–83% defense with strong efficiency. Edge-tier PoA validation adds a median $\approx$ 0.39 ms per partial block for validator sets up to 10 and remains sub-ms with partitioned sub-quorums up to 30, while cloud BFT provides accountable finality off the critical path. In short, EdgePlus delivers a single, coherent scheduling–control–security stack for 6G edge computing, built on a feasibility-first design that preserves reliability, locality, and security at scale.**

***Index Terms*—Blockchain-lite security, constrained reinforcement learning, low-latency networks, multiagent systems, resource allocation, sixth-generation (6G) edge computing.**

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I. Introduction

OUR increasing daily reliance on connected services such as smart transportation, personalized healthcare, and virtual workspaces has stirred demand for more efficient real-time and data-intensive applications. In smart-city air-quality monitoring, millions of battery-powered sensors embedded in lampposts and street furniture generate pollutant readings at sub-second intervals, creating highly variable and bursty data streams. This motivates the emergence of sixth-generation (6G) technologies that seek to provide ultralow latency and massive connectivity across heterogeneous computing devices [1], [2], [3]. Building on fourth and fifth generation (4G/5G) networks, 6G aspires to deliver near-instantaneous responsiveness for billions of distributed devices across different arenas, from city traffic management to rural agricultural monitoring [3], [4], [5], [6].

The evolution toward 6G brings unprecedented device density, heterogeneity, and stringent quality of service targets. Ultra-dense deployments with sub-ms control loops intensify the need for *decentralized*, *constraint-faithful*, and *secure* scheduling. In practice, bursty arrivals, mobility, and energy limits collide with per-slice radio/computational budgets and hard deadlines, while end-to-end confidentiality and integrity must be preserved with minimal latency overhead [1], [5], [7], [8]. Heuristic offloading and unconstrained RL often degrade under these conditions [9], [10], [11], [12], violating deadlines or overusing the cloud, whereas heavyweight consensus can blow real-time budgets [13], [14].

EdgePlus seeks to address these challenges using a feasibility-first, 6G-native design. The inner loop uses cooperative PPO (CCPPO) with a centralized critic, but *samples only from feasible actions* via masks that enforce routing order, slice quotas, capacity/energy limits, and deadline feasibility (see Sections III-C and III-E). A *constricted* particle swarm optimization (PSO) outer loop performs nonlocal exploration and accepts a basin only if SLA-aligned metrics improve (see Section III-G). To shape choices without breaking feasibility, a lightweight advisory layer, Nash sharing [15], [16], Ziegler–Nichols, initialized PID with Jury certification [17], and leader election, adds *bias terms* to logits that remain

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masked (see Sections III-F and III-I). Finally, a two-tier *blockchain-lite* ledger provides sub-ms edge validation proof-of-authority (PoA) with cloud Byzantine fault-tolerant (BFT) finalization [18], and its reputation/debt signals feedback into state, reward, and masks (see Sections III-H and III-M).

We evaluate EdgePlus in a 100 km^2 city-scale simulator (10 clusters) with a deadline-aware workload derived from the Brasov traces across three regimes (50 000/100 000/150 000; Sections IV-B and IV-C). Results are organized per objective (OBJ1–OBJ7; Sections V-A–V-G). Initially, all schedulers achieve $\approx$ 100% success at light load (OBJ2; Section V-B); but under stress, ccppo_edgeplus yields the best reliability, latency, throughput balance (OBJ1; Section V-A); as load scales, feasibility-first admission compresses tails and stabilizes violations (OBJ3; Section V-C). Higher locality tracks lower cloud offload and, in aggregate, fewer misses (OBJ4; Section V-D). Ablations show masks are foundational, advisory/PSO add consistent headroom, and ledger feedback improves placement/tails (OBJ5; Section V-E). The PID operates in a locally robust pocket under $\pm$20% gain drift (OBJ6; Section V-F). Security experiments demonstrate 98%–99% mitigation for baseline confidentiality, integrity, and availability (CIA)/auth vectors with sub-ms to low-ms overhead; PoA remains sub-ms while cloud BFT provides accountable finality (OBJ7; Section V-G).

A. Contributions

This work provides as follows.

1) *Feasibility-First MultiAgent RL for 6G:* Masked CCPPO that enforces routing, slicing, capacity/energy, and deadline feasibility *by construction* (see Sections III-E and III-C).
2) *Integrated Learning–Control Co-Design:* Nash sharing, Jury-certified PID [17], and leader election bias feasible actions without violating masks (see Sections III-F and III-I).
3) *SLA-Guarded Bilevel Optimization:* A constricted PSO outer loop that accepts basins only when SLA metrics improve (see Section III-G).
4) *Blockchain-Lite Feedback With DKMS:* Sub-ms PoA at the edge with cloud BFT finality; reputation/debt integrated into state, reward, and masks for tamper-aware scheduling (see Sections III-H and III-M).
5) *Comprehensive City-Scale Evidence:* Across 50 000–150 000, EdgePlus sustains top reliability under stress, preserves locality with low cloud usage, quantifies component value via ablations, certifies control robustness, and achieves strong security efficiency (see Sections V-A–V-G).

B. Organization

Section II positions our work. Section III formalizes the system model and algorithms. Section IV details datasets, simulator, methods, and key performance indicators (KPIs). Sections V-A–V-G present OBJ1–OBJ7 with the agreed narrative flow. Section VII concludes.

II. Related Work

In the rapidly evolving landscape of 6G edge computing, prior studies provide valuable building blocks but remain fragmented, optimizing scheduling, mobility, or security in isolation, without an integrated, constraint-faithful, and tamper-aware stack. Below we review representative efforts in 1) wireless/edge architectures, 2) multiagent RL and swarm optimization, 3) cyber-physical control, and 4) security for 6G; we then position *EdgePlus* as a unified alternative that couples feasibility-first MARL, advisory control, PSO refinement, and a blockchain-lite feedback loop (formalized in Section III).

A. Current 6G Wireless and Edge Architectures

Frameworks such as ParallEdge [3] and EdgeGO [4] decouple task offloading from mobility, treating them as sequential or parallel but independent modules. ParallEdge's genetic-algorithm clustering incurs $O(N^2)$ overhead and offers no formal latency or deadline guarantees [19], driving unbounded worst case delays as node counts scale. EdgeGO's in-route computing optimizes static trajectories but omits dynamic obstacle avoidance; under bursty, high-density loads, it suffers deadline misses and load imbalance due to rate-based heuristics.

Zhao et al. [20] diversity-aware placement focuses on offline node positioning to raise throughput via spatial diversity, *not* on runtime allocation or mobility handoff; once deployed, node locations are fixed and the system is brittle to shifting hotspots. Vehicle-mounted MEC approaches (e.g., Liu et al. [21]) solve joint path planning and offloading via centralized MILP/branch-and-bound; despite optimality at small scale, repeated NP-hard reoptimization is incompatible with millisecond windows and hard energy/capacity constraints.

Most wireless/edge designs either omit end-to-end security or rely on heavyweight, monolithic blockchains that impose multisecond consensus latencies on commodity edge hardware. Likewise, although control-theoretic elements exist in adjacent domains, they are rarely integrated into edge orchestration.

Rather than a side-by-side experimental table, we summarize the key gaps inline: 1) offline or centralized optimization without hard-constraint enforcement, 2) limited or no runtime mobility/redistribution, and 3) absent or impractically heavy security. *EdgePlus* addresses these by enforcing feasibility at sampling time, adding lightweight advisory control, and integrating ledger-derived reputation/debt into state, reward, and masks.

Recent works advance specific facets but retain these limitations. Lin et al. [22] propose AI-driven access for SAGIN with massive connectivity but no hard constraints or integrated security guarantees. He et al. [23] study multiobjective DRL for satellite time–frequency allocation, balancing throughput/energy yet overlooking mobility and tamper-evidence, and design anti-collision strategies for MF–TDMA ranging that improve massive access without ledgered security or edge mobility adaptation [24]. EdgePlus unifies these strands with masked CCPPO, PSO refinement, and blockchain-lite;

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its reliability/latency advantages are demonstrated across OBJ1–OBJ5 (see Sections V-A, V-C, and V-E).

B. Multiagent Reinforcement Learning and Swarm Strategies

MADRL and swarm methods promise decentralized, adaptive resource management, but common instantiations fall short under bursty loads and mobility. Works by Zhang et al. [25], [26], and Lu et al. [27] reduce latency via distributed policies yet assume largely static graphs; under nonstationarity, learned policies can violate energy budgets or overflow queues. Time-sensitive approaches such as AoI minimization [28] that seek to optimize averages and not hard per-task deadlines usually risk SLA violations.

Coordinated PPO variants (such as CoPPO [29] and CMPPO [30]) introduce interagent communication, but they lack explicit fairness/feasibility enforcement. Thus, without Nash-style sharing [15], [16] and hard masks, agents may offload to suboptimal servers during spikes. Swarm-inspired PSO derivatives [30], [31] accelerate convergence on fixed surfaces but lack runtime safety gates and feedback coupling to handle sudden topology changes.

Hybrid vehicle-MEC solvers (e.g., [21]) mix MILP with search for joint path/offload but do not scale to hundreds of moving agents with ms-level replanning; pure game-theoretic sharing [15], [16] assumes small, static games and cannot react at control-loop rates. In contrast, *EdgePlus* embeds feasibility masks within CCPPO (see Section III-E), biases feasible logits via Nash/PID/leader advisory [16], [17] (see Section III-F), and uses a *constricted* PSO outer loop with SLA-gated acceptance (see Section III-G).

C. Cyber-Physical Systems and PID Control

PID loops are widely used in power/industrial control to counter communication delays [17], yet edge-computing schedulers rarely co-design a utilization controller with learning. Without a feedback controller governing mobile activation and cross-cluster sharing, systems oscillate between overload and under-utilization as demand shifts, undermining latency and energy goals. Gains are seldom tuned in situ, and controllers are not certified robust to component drift or crypto workloads.

At the same time, end-to-end cryptography appears piecemeal: AES–GCM for confidentiality [32], [33], Ed25519 for signatures [34], [35], X25519 for key agreement [36], and SHA3/BLAKE2b hashing [37], but without a distributed key-lifecycle or tamper-evident ledger that operates within edge-latency budgets. Full blockchains (PoW/PoS/PoA) can add multisecond delays [18], [33]; DHT-style overlays reduce cost but forfeit strong finality. *EdgePlus* co-designs a Ziegler–Nichols-initialized, Jury-certified PID (see Section III-I) with masked MARL and a two-tier ledger (see Section III-M), ensuring that control stability and crypto overheads remain within SLA bounds.

D. Security in 6G Environments

Zhao et al. [20] disperse edge nodes to maximize coverage/throughput via genetic optimization, but omit runtime redistribution and any cryptographic safeguards; data pipelines remain exposed to interception, spoofing, and undetected manipulation. Liu et al. [21] jointly optimize trajectories/offloading with centralized MILP, achieving latency gains at a small scale while treating deadlines/energy as soft penalties; the threat model assumes full trust and lacks ledgered integrity.

TABLE I
MAPPING OF 6G SECURITY CHALLENGES TO EDGEPLUS MITIGATION STRATEGIES

6G Security Challenge	EdgePlus Mitigation Strategy
Eavesdropping and interception	encryption for confidentiality (Eq. (26a))
Spoofing and impersonation	Ed25519 signatures for authentication (Eq. (25))
Data tampering and integrity	Hybrid hashing (SHA3–256 and BLAKE2b) for integrity checks (Eq. (20))
AI-driven adaptive attacks	Distributed Key Management System (DKMS) for rotation/revocation (Eqs. (27b)–(27c))
Consensus latency at the edge	Two-tier blockchain-lite with PoA validation and cloud BFT [18] (Eqs. (22)–(23))

1) *EdgePlus:* To address these gaps, *EdgePlus* secures ingest/control with X25519 key agreement and AES–GCM AEAD [32] (26a), authenticates blocks/control messages with Ed25519 [34], [35] (25), and logs via a SHA3–256 →BLAKE2b hash pipeline (20) [37]. A two-tier ledger validates *partial* blocks at the edge via PoA with sub-ms overhead [median ≈ 0.39 ms; (22)] and finalizes periodically with cloud BFT (23). A distributed key management system (DKMS) handles key generation/rotation/revocation with risk-aware triggers [38] (27a)–(27c). Table I maps 6G security challenges to these mitigations; OBJ7 reports 98%–99% defense for baseline CIA/auth with ms/sub-ms cost (see Section V-G).
2) *Scope (OBJ7):* Consistent with the *Security Framework* in Section III-M, our evaluation focuses on confidentiality, integrity, authentication, ledger tamper-evidence, and DKMS behavior; purely physical-layer availability attacks are out of scope.

Collectively, these threads motivate a unified approach. Prior systems typically optimize *either* scheduling, *or* mobility, *or* security; *EdgePlus* integrates all three with feasibility-first MARL, advisory control, and ledger-backed feedback—formalized in Section III and validated across OBJ1–OBJ7 (see Sections V-A–V-G).

III. EDGEPLUS SYSTEM MODEL

A. System Model and Notation (6G-Aware)

Notation is defined on first use and KPIs are summarized in Table V. To start we denoted our clusters as $\mathcal{C} = \{1, \ldots, C\}$ and servers $\mathcal{S} = \mathcal{S}^{\text{stat}} \cup \mathcal{S}^{\text{mob}} \cup \mathcal{S}^{\text{cloud}}$ with $c(s) \in \mathcal{C}$ ($c(s) = 0$ for cloud). Tasks $\mathcal{T} = \{1, \ldots, N\}$ arrive over $[0, H]$; task i

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TABLE II
SUMMARY OF EDGEPLUS COMPLEXITY (CONCISE)

Component	Complexity	Dominant factors
CCPPO (train step)	$\mathcal{O}(AD + JA)$	agents A; state/action dims D; constraints J
Constricted PSO	$\mathcal{O}(PI + PM)$	particles P; iterations I; fitness batches M
Sharing (Nash)	$\mathcal{O}(C^2)$	clusters C per round
Mobility (CBS–D* Lite)	$\mathcal{O}(TV)$	relocations T; graph nodes V
Security path	$\mathcal{O}(L + T_{\text{tx}})$	payload bytes L; tuples per block T_{tx}

TABLE III
TASK FEATURE SCHEMA AND TRANSFORMATIONS

Field	Unit/Type	Source	Transform / Rationale
t_i	s (epoch)	timestamp	timezone normalize; enforce monotonicity per site
b_i	bits	payload size	encode pollutant vector + metadata; gzip estimate
k_i	cycles	proc. cost	per-model lookup (light CNN) ⇒ cycles via FLOPs/f_s
m_i	MB	mem. footprint	static model+buffer budget for class p_i
p_i	URLLC, eMBB, mMTC	tier map	WHO-driven tier→slice (Alg. 3)
d_i	s (deadline)	tier map	SLA per tier; slack sampled within tier band
loc_i	lat/long	site metadata	project to (x, y); cluster id via weighted K-means
tier_i	Immediate, Routine, Delayed	policy table	thresholds + trends (robust z-scores and gradient flags)
prio_i	High, Moderate, Low	policy table	break ties by exceedance magnitude with hysteresis

TABLE IV
EPISODE CONSTRUCTION AND CLASS SHARES

Regime	Total tasks	URLLC (%)	eMBB (%)	mMTC (%)
Light	50,000	20	50	30
Medium	100,000	20	50	30
Stress	150,000	20	50	30

has $(t_i, d_i, c(i), p_i, k_i, m_i, b_i)$ with $p_i \in \{\text{URLLC}, \text{eMBB}, \text{mMTC}\}$ (URLLC: ultrareliable low-latency communications; eMBB: enhanced mobile broadband; mMTC: massive machine-type communications; weights $w_{p_i} > 0$). Each server s has rate f_s (cycles/s), memory M_s, (mobiles) energy $E_s^{\max}$, load $\rho_s \in [0, 1]$, backlog Q_s (sec). Each cluster c supports 6G slices $k \in \{\text{URLLC}, \text{eMBB}, \text{mMTC}\}$ with reserved capacities $\{\Gamma_{c,k}^{\text{radio}}, \Gamma_{c,k}^{\text{compute}}\}$.

TABLE V
KPIS USED ACROSS OBJ1–OBJ7

Symbol	Name	Definition / Notes
σ	Success rate	Completed on-time / total (%)
Viol.%	Violation rate	Missed deadlines / total (%)
L_{p50}, L_{p95}	Latency percentiles	End-to-end latency (s) incl. tx/move/queue/proc
Throughput	Throughput	Completed tasks per second
Energy	Total energy	$\sum_i E_{i,s(i)}$ over episode
LI	Locality Index	% tasks executed in home cluster $c(i)$
χ_{cloud}	Cloud-offload ratio	As in Sec. III-D
URLLC tail rel.	URLLC tail reliability	$\widehat{\Pr}[L_{i,s(i)} \le d_i \mid p_i{=}\text{URLLC}]$
$\mathcal{J}$	Fairness (Jain's)	$\frac{(\sum_c \Theta_c)^2}{C \sum_c \Theta_c^2}$ over clusters
SEI	Security Efficiency	Defense% / median overhead (ms)
CAF	Consensus Amplif. Factor	Ledger work / base work

B. Latency and Energy

Assigning task i to server s yields

$$L_{i,s} = L_{i,s}^{\text{tx}} + L_{i,s}^{\text{move}} + L_{i,s}^{\text{queue}} + L_{i,s}^{\text{proc}}. \tag{1}$$

We use $L_{i,s}^{\text{tx}} = \eta_{\text{OH}} b_i / \bar{R}_{i,s}$ (effective 6G rate $\bar{R}_{i,s}$ captures link/RIS/backhaul), $L_{i,s}^{\text{move}} = 0$ for stationary/cloud and $L_{i,s}^{\text{move}} = \|x_i^\star - x_s(t_i)\|/v_s$ for mobiles, $L_{i,s}^{\text{proc}} = k_i/f_s$, and $L_{i,s}^{\text{queue}}$ from a calibrated queue surrogate. Energy

$$E_{i,s} = E_{i,s}^{\text{tx}} + E_{i,s}^{\text{move}} + E_{i,s}^{\text{comp}} \tag{2}$$

e.g., $E_{i,s}^{\text{tx}} = P_{i,s}^{\text{tx}} L_{i,s}^{\text{tx}}$, $E_{i,s}^{\text{move}} = \chi_s \|x_i^\star - x_s(t_i)\|$, $E_{i,s}^{\text{comp}} = \kappa_s k_i$.

C. Feasibility, Routing Order, and 6G Slicing Constraints

1) Order: Stationary in $c(i)$ → mobile in $c(i)$ → mobile in $\mathcal{N}(c(i))$ → cloud. Let $M_{i,s} \in \{0, 1\}$ be routing + resource feasibility, $x_{i,s} \in \{0, 1\}$ the assignment

$$\begin{aligned} &\sum_{s \in \mathcal{S}} x_{i,s} = 1, \quad x_{i,s} \le M_{i,s} \ \forall i, s \\ &t_i + L_{i,s} \le d_i + U_i \left(1 - x_{i,s}\right) \quad \forall i, s \\ &\sum_i x_{i,s} \tfrac{k_i}{f_s} \le H \quad \forall s \ \sum_i x_{i,s} E_{i,s} \le E_s^{\max} \ \forall s \in \mathcal{S}^{\text{mob}}. \end{aligned} \tag{3}$$

Big-M term. Here U_i is a standard big-M slack (large enough to deactivate the deadline constraint when $x_{i,s} = 0$); it does not affect feasibility when $x_{i,s} = 1$.

Per-slice reservations in cluster c (6G slicing)

$$\begin{aligned} \sum_{i:\, p_i = k} \ \sum_{s:\, c(s) = c} x_{i,s} \tfrac{b_i}{\bar{R}_{i,s}} &\le \Gamma_{c,k}^{\text{radio}} \\ \sum_{i:\, p_i = k} \ \sum_{s:\, c(s) = c} x_{i,s} \tfrac{k_i}{f_s} &\le \Gamma_{c,k}^{\text{compute}}. \end{aligned} \tag{4}$$

D. Objective, Cloud-Offload KPI, and URLLC Tail

The multiobjective is

$$J(x) = \sum_i \left[\alpha\, w_{p_i} \mathbf{1}\left(t_i + L_{i,s(i)} > d_i\right) + \beta\, w_{p_i} L_{i,s(i)}\right.$$

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$$+ \gamma E_{i,s(i)} + \delta \mathbf{1}\left(s(i) \in \mathcal{S}^{\text{cloud}}\right)\Big]. \tag{5}$$

Cloud-migration (offload) ratio $\chi_{\text{cloud}} = 1/|\mathcal{I}_{\text{done}}| \sum_i \mathbf{1}[s(i) \in \mathcal{S}^{\text{cloud}}]$. For URLLC tasks, enforce tail reliability

$$\widehat{\Pr}\left[L_{i,s(i)} \le d_i\right] \ \ge \ 1 - \varepsilon_{\text{URLLC}}. \tag{6}$$

Online-shaped reward

$$\begin{aligned} r_t = &+\eta_1 w_{p_i} \mathbf{1}[\text{on-time}] - \eta_2 w_{p_i} \mathbf{1}[\text{late}] \\ &- \eta_3 \tilde{L}_t - \eta_4 \tilde{E}_t - \eta_5 \mathbf{1}\left[a_t \in \mathcal{S}^{\text{cloud}}\right]. \end{aligned} \tag{7}$$

E. Masked Markov Decision Process (MDP) and CCPPO (CCPPO With Centralized Critic)

We use a cooperative multiactor policy with a centralized critic. At step t, the agent observes $\mathbf{s}_t$; feasibility masks enforce $\pi_\theta(a = s \mid \mathbf{s}_t) = 0$ if $M_{i,s} = 0$. Actors $\{\pi_{\theta_j}\}$ cooperate via a shared value function $V_\psi(\mathbf{s}_t)$. With generalized advantage estimation (GAE) [11], [39] $\hat{A}_t$ and ratio ρ_t

$$\begin{aligned} \mathcal{L}^{\text{actor}} = &-\mathbb{E}\Big[\min\left(\rho_t \hat{A}_t, \ \text{clip}\left(\rho_t, 1-\epsilon, 1+\epsilon\right)\hat{A}_t\right)\Big] \\ &+ \lambda_{\text{KL}} \ \text{KL}\left(\pi_\theta^{\text{old}} \| \pi_\theta\right) - \lambda_{\mathcal{H}} \mathbb{E}\left[\mathcal{H}(\pi_\theta)\right] \end{aligned} \tag{8}$$

and $\mathcal{L}^{\text{critic}} = \mathbb{E}[(V_\psi(\mathbf{s}_t) - \hat{V}_t)^2]$.

F. Advisory Coordination (Nash/PID/Leader) and Masked Softmax

Let raw logits be z_s and advisory-augmented logits

$$\tilde{z}_s = z_s + \lambda_N \log s_{c(s)} + \lambda_P u_{c(s)} + \lambda_L \mathbf{1}\left[s = s_{c(s)}^{\text{lead}}\right]. \tag{9}$$

Masked softmax

$$\pi_\theta(a = s \mid \mathbf{s}) = \frac{M_{i,s}\, e^{\tilde{z}_s}}{\sum_{j \in \mathcal{S}} M_{i,j}\, e^{\tilde{z}_j}}. \tag{10}$$

Nash selects $s_c \in [s_{\min}, s_{\max}]$ by best response to maximize [15], [16] $u_c = -\alpha_N s_c \,\text{load}_c - \beta_N \,\text{energy}_c - \gamma_N (s_c - 1)^2$ (stop when $\|s^{(t+1)} - s^{(t)}\|_\infty < \varepsilon$). PID uses $e_c = \rho_c - \rho_{\text{set}}$ [17]

$$\begin{aligned} u_c(t) = &K_p e_c(t) + K_i \int e_c \, dt \\ &+ K_d \dot{e}_c(t), \quad u_c \in [-1, 1]. \end{aligned} \tag{11}$$

Leader: $\text{score}_s = \omega_1(1 - \text{load_ratio}_s) + \omega_2 \,\text{energy_ratio}_s$; choose $s_c^{\text{lead}} = \arg\max \text{score}_s$.

G. Bilevel Outer Loop and PSO Escaping Local Optima

Outer parameters Θ (inits/regularizers/advisory gains) are tuned by PSO; inner loop learns $\theta^\star(\Theta)$

$$\begin{aligned} &\max_{\Theta \in \mathcal{B}} \quad \mathcal{F}(\Theta) = \mathbb{E}\left[J\left(\pi_{\theta^\star(\Theta)}\right)\right] \\ &\text{s.t.} \quad \theta^\star(\Theta) \in \arg\max_\theta \ \mathbb{E}\left[\sum_t \gamma^t r_t\right] \ \text{(CCPPO with masks)}. \end{aligned} \tag{12}$$

Constricted PSO performs nonlocal basin-hopping using masked episodic returns

$$w_t = w_{\max} - (w_{\max} - w_{\min}) \ \frac{t}{I} \tag{13a}$$

$$\mathbf{v}_{t+1}^{(i)} = k\, w_t\, \mathbf{v}_t^{(i)} + k\, c_1\, \mathbf{r}_1 \odot \left(\mathbf{p}^{(i)} - \boldsymbol{\theta}_t^{(i)}\right) + k\, c_2\, \mathbf{r}_2 \odot \left(\mathbf{g} - \boldsymbol{\theta}_t^{(i)}\right) \tag{13b}$$

$$\boldsymbol{\theta}_{t+1}^{(i)} = \boldsymbol{\theta}_t^{(i)} + \mathbf{v}_{t+1}^{(i)} \tag{13c}$$

$$k = \frac{2}{\left|2 - \phi - \sqrt{\phi^2 - 4\phi}\right|}, \quad \phi = c_1 + c_2 > 4. \tag{13d}$$

Acceptance constraints enforce SLA-aligned safety: adopt a PSO basin only if success improves and ($\chi_{\text{cloud}} \le 0.07$, fail $\le$ 0.05, stationary ≥ 0.80) are respected.

H. Blockchain-Backed Feedback: State, Reward, and Masks

From the ledger $\mathcal{L}_t$ compute reputation and debt

$$\begin{aligned} \rho_s^{\text{rep}} &= \text{EMA}(\mathbf{1}[\text{on-time}] - \kappa\, \mathbf{1}[\text{SLA violation}]) \\ d_c &= \max\left\{0, \ \bar{\Theta} - \Theta_c\right\}. \end{aligned} \tag{14}$$

Integrate systemically: augment state with (ρ^{rep}, d); shape

$$r_t \leftarrow r_t - \eta_6\, \phi\left(\rho_{a_t}^{\text{rep}}\right) - \eta_7\, d_{c(a_t)}. \tag{15}$$

Shaping map. We use a hinge penalty on low reputation

$$\phi(\rho) = \max\{0, \ \rho_{\text{th}} - \rho\}, \quad \rho \in [0, 1] \tag{16}$$

with $\rho_{\text{th}} \in (0, 1)$ (default 0.6). For smoothness, a softplus variant $\phi_\beta(\rho) = 1/\beta \log\left(1 + e^{\beta(\rho_{\text{th}} - \rho)}\right)$ is interchangeable (default β=10).

For URLLC, low-reputation servers can be quarantined by setting $M_{i,s}$=0 whenever $\rho_s^{\text{rep}} < \rho_{\text{th}}^{\text{URLLC}}$ (with $\rho_{\text{th}}^{\text{URLLC}} \ge \rho_{\text{th}}$), preserving mask invariants (coherent with routing and slices).

I. PID Robustness Under ±20% Gain DriftPID Robustness Under ±20% Gain Drift

Approximate utilization by $G(s) = K/(\tau s + 1)e^{-Ls}$; discretize at T_s to $G(z)$. With discrete PID $C(z)$, closed-loop roots of $P(z) = 1 + C(z)G(z)$ must lie inside the unit disk. Robustness set

$$\begin{aligned} \mathcal{K} = \Big\{&\left(K_p', K_i', K_d'\right): \\ &K_p' = \left(1+\delta_p\right)K_p, \quad K_i' = (1+\delta_i)K_i \\ &K_d'\,(1+\delta_d)K_d, \ |\delta_\cdot| \le 0.2\Big\}. \end{aligned} \tag{17}$$

Gains are initialized via Ziegler–Nichols, then certified by Jury's test over $\mathcal{K}$; if any corner violates (M_p, T_s) or exhibits oscillation under temperature-induced drift, we reduce K_p or raise ρ_{set} and retest all corners.

J. Module Interactions and Dataflow (Smart-City Context)

DataGenerator→ CSVs (tasks/servers/topology, slice tags). Scheduler → Env (routing masks, slice quotas, and reward). CCPPO → masked cooperative policy with a centralized critic (cluster-aware actors). Advisory (Nash/PID/Leader) → bias logits, never break masks. Blockchain → reputation/debt enter state, reward, and masks. PSO → pretune/interleave around CCPPO; adopt a basin only if masked episodic return *and* SLA metrics improve. Evaluator → KPIs: success, $p50/p95$ latency, energy, χ_{cloud}, fairness, and URLLC violations.

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Algorithm 1 EdgePlus Training With CCPPO, Advisory Bias, and PSO

Input: CSVs; graph; servers; $(\gamma, \epsilon, \lambda_{\mathcal{H}}, \lambda_{\mathrm{KL}})$; advisory $(\lambda_N, \lambda_P, \lambda_L)$; PID $(K_p, K_i, K_d, \rho_{\mathrm{set}})$; PSO $(P, I, w_{\max}, w_{\min}, c_1, c_2, T_{\max})$.
Build Env; init $\{\pi_{\theta_j}\}$ and centralized V_ψ; init Nash/PID/Leader.
if PSO pre-tune **then**
    Run I PSO iters on (θ, ψ) with masked rollouts; warm-start to best fitness.
for episode = $1, 2, \ldots$ **do**
    Reset env and buffer $\mathcal{B}$.
    while not done **do**
        Observe $\mathbf{s}_t$; build mask $M_{i,\cdot}$ by routing+resources.
        Compute $s_c, u_c, s_c^{\mathrm{lead}}$; form $\tilde{z}$ and sample a_t from masked softmax.
        Step env; log $(\mathbf{s}_t, a_t, r_t, \mathbf{s}_{t+1}, M_{i,\cdot}) \in \mathcal{B}$; update ledger-derived (ρ^{rep}, d) and shape r_t; optionally quarantine poor-reputation servers for high-priority tasks.
    PPO update (clipped); update centralized critic; apply entropy and KL penalties; clip gradients.
    if episode mod K_{PSO}=0 and PSO interleave **then**
        Run a few PSO steps around (θ, ψ); accept basin only if success improves and $(\chi_{\mathrm{cloud}}, \mathrm{fail}, \mathrm{stationary})$ meet SLA thresholds.
    Persist trace_epXX.csv and training metrics; run evaluator for KPIs.

Algorithm 2 Evaluation and PID Robustness (URLLC Emphasis)

Freeze (θ, ψ).
Evaluation loop
for $e = 1$ to E_{eval} **do**
    Run masked policy with advisory; record episode trace.
Compute KPIs
σ (success), Θ (throughput), L_{p50}, L_{p95}, $\bar{E}$ (mean energy), χ_{cloud} (cloud share), $\mathcal{J}$ (Jain fairness), $\hat{\varepsilon}$ (URLLC violation).
PID robustness
Discretize $G(s)$ at sampling T_s to obtain $G(z)$.
Form $P(z) = 1 + C(z)G(z)$ for the nominal gains and all $\pm 20\%$ corners in $\mathcal{K}$.
for each corner $(K_p', K_i', K_d') \in \mathcal{K}$ **do**
    Apply Jury's test to $P(z)$; record stability and (M_p, T_s) limits.
if any corner unstable or (M_p, T_s) limits exceeded **then**
    Reduce K_p or increase ρ_{set}; re-test all corners.

K. Design Invariance

Masks enforce feasibility invariantly; advisory only reorders feasible actions. PSO is a bilevel outer program over CCPPO parameters and adopts a basin only when masked episodic returns and SLA-constrained shares improve, ensuring principled nonlocal exploration. Blockchain-derived reputation/debt enter state, reward, and masks, three coherent interfaces, while PID gains are certified robust over $\pm 20\%$ drift.

L. End-to-End Algorithms (Short-Line, Column-Safe)

To make the end-to-end pipeline reproducible, we summarize the full workflow in compact, column-safe pseudocode. Algorithm 1 presents the EdgePlus training loop (masked CCPPO with advisory biasing and an optional SLA-gated PSO outer loop). Algorithm 2 presents the common evaluation protocol used across OBJ1–OBJ7. Algorithm 3 provides the deterministic labeling and deadline assignment procedure used to build the deadline-aware workload (Section IV-B).

M. Security Framework

To secure data end-to-end without violating edge deadlines, *EdgePlus* implements a two-tier *blockchain-lite* ledger with hybrid cryptography. As indicated in the system model (see Section III-A), per-cluster logging at the edge ($c \in \mathcal{C}$) feeds runtime *reputation* and *debt* signals into the scheduler state/reward/masks (see Section III-H), while a cloud tier provides eventual global immutability. Unlike monolithic blockchains, our ledger is intentionally lightweight so as not to violate millisecond-level URLLC deadlines.

1) Tunable Adversary Budget: The adversary's compute share λ_{adv} (see Section V-G) is a configurable knob. To understand resilience across attack intensities, we evaluate multiple values (0.05, 0.10, and 0.25) rather than fixing λ_{adv} to a single default.

2) Task Log Record and Partial Blocks (Edge Tier): For each task i executed on server $s(i)$, cluster c emits a compact log tuple

$$\ell_{i,c} = \big(i,\ c,\ s(i),\ p_i,\ d_i,\ t_i,\ t_i^{\mathrm{end}},\ \mathbf{1}[\text{on-time}]\ E_{i,s(i)},\ \mathrm{nonce}_i,\ \mathrm{tag}_i\big) \tag{18}$$

where tag_i is the AES–GCM tag (26a). Per-message authenticity is provided by AES–GCM; to minimize overhead, device-level signatures are not attached to each tuple [32], [33]. Instead, the authority signs each full block (25), giving *block-level nonrepudiation*. Identifiers are pseudonymized, and access is enforced via cluster-scoped ACLs.

At the local edge, recent tuples are batched into a *partial block* that links to the previous hash

$$\mathrm{Block}_{c,r}^{(\mathrm{local})} = \{\ell_{i,c}\}_{i \in \mathcal{B}_r} \parallel h_{c,r-1} \tag{19}$$

$$h_{c,r} = \mathrm{BLAKE2b}\big(\mathrm{SHA3\text{-}256}\big(\mathrm{Block}_{c,r}^{(\mathrm{local})}\big)\big) \tag{20}$$

forming a tamper-evident chain per cluster. To match runtime load, each cluster adapts its local block size using its utilization $\rho_c(t)$ and write rate $\lambda_c(t)$

$$S_c(t) = S_{\min} + \left(\frac{\rho_c(t)}{\rho_{\max}} + \frac{\lambda_c(t)}{\lambda_{\max}}\right)(S_{\max} - S_{\min}). \tag{21}$$

3) Local Validation (PoA) and Global Finalization (BFT): Local consensus uses lightweight PoA [33] among cluster authorities $\mathcal{A}_c$. Each proposed partial block is checked with bounded cost proportional to $(|\mathcal{A}_c| + |\mathcal{B}_r|)$

$$\mathrm{Validate}\big(\mathrm{Block}_{c,r}^{(\mathrm{local})},\ \mathrm{Auth}_j\big) = \mathbf{1}\big(\mathrm{Auth}_j \in \mathcal{A}_c \wedge \mathrm{Verify}\big(\mathrm{Block}_{c,r}^{(\mathrm{local})}\big)\big). \tag{22}$$

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where Verify(·) checks hashes, AES–GCM tags, and any required authority signatures. On our testbed, this path is sub-millisecond (median ≈ 0.39 ms) for validator sets $|\mathcal{A}_c| \le 10$ and typical block sizes. We calibrate the PoA protocol [18] for larger validator sets ($|\mathcal{A}_c| \le 30$) by partitioning validators into sub-quorums of at most ten nodes and rotating leaders; median overhead remains sub-ms up to $|\mathcal{A}_c| = 30$. The ledger includes fork detection: if network partitions produce competing partial blocks, each fork is tagged with a unique ID; upon reconnection, the protocol reconciles the longest valid chain and marks conflicting forks as invalid, reducing the reputation of the offending authorities. Periodically, clusters will forward $h_{c,r}$ to the cloud tier, where immutability is finalized using a BFT protocol [18]

$$\text{Block}_{c,r}^{(\text{global})} = \text{Reinforced_Consensus}\left(\text{Block}_{c,r}^{(\text{local})}\right). \quad (23)$$

4) Block Aggregation, Checks, and Signature: At each global round r, the BFT primary aggregates cluster hashes in ascending c

$$\text{FullBlock}_r = \left(h_{1,r} \parallel h_{2,r} \parallel \cdots \parallel h_{C,r}\right) \quad (24)$$

$$\sigma_r = \text{Sign}_{\text{Ed25519}}(\text{FullBlock}_r). \quad (25)$$

Receivers verify that $|\text{FullBlock}_r| = C$ and cluster IDs are strictly increasing, accept σ_r, and append; leaders then recompute their local $h_{c,r}$ (20) and append upon success. False-accept probability is negligible: no invalid block was accepted in our evaluation, yielding a rule-of-three upper bound of $< 1/N$ (with N the number of evaluation episodes).

5) Session Setup and Authenticated Encryption: All IoT→edge and internode channels derive per-session symmetric keys with X25519 ECDH; payloads and control messages are protected using AES–GCM (confidentiality + integrity) [32] and Ed25519 (entity authentication) [34], [35]

$$\left(\text{cipher, tag}\right) = \text{AES-GCM}_k\left(\text{msg, IV}\right) \quad (26a)$$

$$k = X25519\left(sk_A, pk_B\right) \quad (26b)$$

$$k = X25519\left(sk_B, pk_A\right). \quad (26c)$$

Each session enforces nonce (IV) uniqueness per key using a 96-bit IV constructed from a per-session salt and a monotonic counter.

6) Key Lifecycle (DKMS) and Triggers: A distributed KMS manages key generation, rotation, and revocation across nodes $N=\{n_1, \ldots, n_m\}$ [38]

$$\left(sk_i,\ pk_i\right) = G(n_i) \quad (27a)$$

$$\left(sk_i',\ pk_i'\right) = R(sk_i,\ pk_i,\ t) \quad (27b)$$

$$\text{Revoke}(n_i):\ \left(sk_i,\ pk_i\right) \mapsto \text{Revoked}. \quad (27c)$$

Rotations are time-based with *risk-aware* acceleration: if ledger-derived reputation in a cluster drops below a threshold *or* cryptographic anomalies are detected (e.g., replayed signatures or hash mismatches), the DKMS shortens rekey intervals for affected links and can trigger immediate revocation [38]. To avoid rotation-induced DoS, we enforce a floor and hysteresis: rekey interval $\ge T_{\min}$, back-off on repeated triggers, and require reputation to remain below threshold for N consecutive blocks. The DKMS is partition-tolerant: if the network splits, each partition continues to operate with its latest keys; upon reconnection, rotations and revocations are reconciled via the global ledger, and any compromised keys are invalidated.

7) Ledger → Scheduler Hooks (Coherence With Masks/Reward): Every committed $\ell_{i,c}$ updates per-server reputation and cluster debt exactly as used by the scheduler in Section III-H

$$\rho_s^{\text{rep}} \leftarrow \text{EMA}(\mathbf{1}\,[\text{on-time}] - \kappa\,\mathbf{1}\,[\text{SLA violation}])$$
$$d_c \leftarrow \max\{0,\ \bar{\Theta} - \Theta_c\}. \quad (28)$$

This feedback as features and shape the reward

$$r_t \leftarrow r_t - \eta_6\,\phi\left(\rho_{a_t}^{\text{rep}}\right) - \eta_7\,d_{c(a_t)} \quad (29)$$

and, for URLLC tasks, can *tighten the feasibility mask* by quarantining low-ρ^{rep} servers (set $M_{i,s} = 0$ under the policy in Section III-H). The hook layer also detects reputation-manipulation attempts: if a cluster repeatedly reports spurious completions or deadline violations, its reputation is reduced, and the masks quarantine its servers.

8) Scope: Our design preserves *confidentiality* and *integrity*, via X25519 +AES–GCM and Ed25519,and provides ledger tamper-evidence and key lifecycle management. It does not restore radio availability or mitigate physical-layer availability attacks. Cryptographic overhead remains small relative to radio duty cycle; we evaluate the impact of security settings on scheduling performance in Section V-G.

N. System Integration and Workflow

EdgePlus integrates cryptography, logging, and scheduling as follows (see Fig. 1).

1) *Secure Ingest:* Devices derive session keys via X25519 and protect payloads with AES–GCM (26a) and (26b); control-plane messages are authenticated with Ed25519.
2) *Edge Logging:* Leaders roll task tuples into partial blocks with PoA validation and adaptive sizing (19)–(21); this path is sub-ms (see Section III-M).
3) *Fork Handling:* If network partitions yield competing partial blocks, fork IDs are assigned and reconciled upon reconnection; conflicting blocks reduce the reputation of the offending cluster.
4) *Ledger Hooks:* ρ^{rep} and d_c update immediately (see Section III-H), feeding state, reward and masks; the shaping function $\phi(\cdot)$ is defined once in Section III-H.
5) *Scheduling:* Masked CCPPO selects feasible actions; advisory (Nash/PID/Leader) biases logits without breaking masks (see Sections III-E and III-F).
6) *Adaptation:* On overload, share across clusters, and reroute mobile servers (CBS–D*Lite); constricted PSO refines around feasible basins (see Sections III-G).
7) *Key Lifecycle:* DKMS rotates/revokes with hysteresis/floors (27a)–(27c); rotation triggers can be risk-accelerated by ledger reputation or cryptographic anomalies. Compromised keys cause immediate revocation and global ledger reconciliation once the partition heals.
8) *Finalization:* Clusters periodically forward $h_{c,r}$ to the cloud for BFT finalization (23), which ensures eventual global immutability.

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Legend: N tasks per window; A agents; D feature dimension; J active constraints; P particles; I PSO iterations; M fitness-evaluation batches; C clusters; T replans/relocations; V graph nodes; L payload size (bytes); T_{tx} logged tuples per partial block.

IV. Experimental Setup

We evaluate *EdgePlus* in a city-scale simulator (see Section IV-C) under three load regimes 50 000, 100 000, and 150 000. For each regime, we run five episodes with identical topology, slice budgets, and RNG seeds. Evaluation uses *policies frozen at inference* (see Section IV-E); the PSO outer loop is disabled during evaluation. Each scheduler acts under its standard decision rule; for example, ddpg samples from a feasibility-masked softmax (renormalized over feasible $M_{i,\cdot}$=1 when masks are enabled), whereas ccppo_lagr applies an unconstrained/Lagrangian policy without masks. The advisory layer and PSO are exercised only when enabled by the configuration, and feasibility masks are toggled per configuration.

We compare *five* schedulers via a common API (see Section IV-D) and run ablations that independently switch feasibility masks, the advisory layer (Nash/PID/Leader), the PSO outer loop, and ledger feedback (see Section V-E). Primary endpoints, success σ, latency percentiles L_{p50}/L_{p95}, locality index (LI), energy, and cloud offload, are defined in Section IV-F. The security scope (OBJ7) covers confidentiality, integrity, authentication, ledger tamper-evidence, and DKMS behavior (see Sections III-M and V-G); physical-layer-only availability attacks are out of scope.

Unless noted otherwise, we report means with 95% confidence intervals across episodes and use Wilcoxon signed-rank tests with Benjamini–Hochberg correction (α = 0.05); rank-biserial r is reported for effect size. For comparative objectives, we prioritize a higher σ and require a noninferior L_{p95} as a tie-breaker.

A. Evaluation Objectives and Protocol

We evaluate seven objectives (OBJ1–OBJ7) that mirror the system design in Section III. This section records only shared experimental conditions (datasets, topology, schedulers, training/inference budget, and KPIs in Table V). For each objective, the *definition*, *metrics*, *experiment*, and *results* are presented narratively in Section V within the corresponding OBJ section, reusing the same figures and tables.

B. Datasets and Preprocessing

We build a deadline-aware, 6G-aligned workload from the *City of Brasov* air-pollution traces (449 sites; August–October 2014).[1] Raw series are transformed into IoT task records with explicit deadlines, time-sensitivity tiers, and priorities, following WHO-aligned handling guidance and our public processing guidelines.[2] This converts passive sensing into an actionable task-allocation benchmark for EdgePlus.

[1]Modified Brasov *Dataset* (ZIP archive, GitHub snapshot.).

[2]Guidelines for Pollutant Processing and Public Health.

The dataset contains *no* embedded attack traces. Security experiments in OBJ7 inject *protocol-level* events at simulation time (CIAA, ledger tamper-evidence, DKMS behavior).

1) Sources and Schema: We use: 1) pollutant time series per site; 2) site metadata (lat/long, altitude); 3) calendar features (hour, weekday). Each row becomes a *task* with fields $\{t_i, b_i, k_i, m_i, p_i, d_i, \text{loc}_i, \text{tier}_i, \text{prio}_i\}$.

2) Label Synthesis: Deadlines, Tiers, Priorities: Deadlines and service classes encode operational urgency and map to 6G slices.

1) Tier →Slice: *Immediate*⇒URLLC, *Routine*⇒eMBB, *Delayed*⇒mMTC.
2) Deadlines: URLLC: $\mathcal{U}[5, 15]$ s; eMBB: $\mathcal{U}[30, 120]$ s; mMTC: $\mathcal{U}[120, 600]$ s.
3) Priorities: High/moderate/low from exceedance magnitude and derivative (spike) with hysteresis to avoid flapping.

This yields $(p_i, d_i, \text{prio}_i)$ consistent with Sections III-C and III-D.

3) Scaling Regimes and Episode Construction: We instantiate three total sizes (50 000/100 000/150 000) while *preserving perepisode shares* (see Table IV). For each size we fix:

1) identical topology, radio/compute slice budgets, and server inventories;
2) identical RNG seeds for sampling deadlines/slack and for device placement jitter;
3) identical training/eval splits across schedulers (policies frozen at eval; see Section IV-E).

4) Feature Engineering for the RL State: Per decision, the observation concatenates: local queue depths and headroom (Q_s, ρ_s, M_s), link rate estimates $\bar{R}_{i,s}$ and RTTs, deadline slack $(d_i - t)$, slice availability $\{\Gamma_{c,k}\}$, neighbor congestion summaries, and ledger signals (ρ^{rep}, d_c). All continuous features are standardized per-split (mean/variance from training only). No adversary/attack ground-truth flags are included in the state; detection is performed by runtime mechanisms, not labels baked into the dataset.

5) Cleaning, Imputation, and Validation: Preprocessing converts the raw pollutant time series into deadline-aware tasks using the following steps:

1) We drop exact duplicates; reconcile clock skew by sitewise monotonic repair.
2) We winsorize extreme sensor spikes at p99.5 (kept for tiering via robust median/IQR).
3) We impute gaps ≤ 10 min via linear interpolation; longer gaps are masked out.
4) We unit-normalize pollutants; check invariants (nonnegative payload, deadline > arrival).
5) We run a static feasibility audit: masks $M_{i,\cdot}$ nonall-zero for >99.9% of tasks.

6) Ablation-Ready Labels: To support OBJ5, we export toggles per task/run for: 1) mask enforcement, 2) advisory bias, 3) PSO outer loop, and 4) ledger feedback injection. Each toggle is logged in the episode header for exact reproduction.

7) Reproducibility: We fix seeds {*38, 39, 40*} for 1) sampling of (d_i, p_i); 2) clusterer initialization; 3) simulator randomness. We publish CSVs: *tasks.csv*, *servers.csv*, *topol-*

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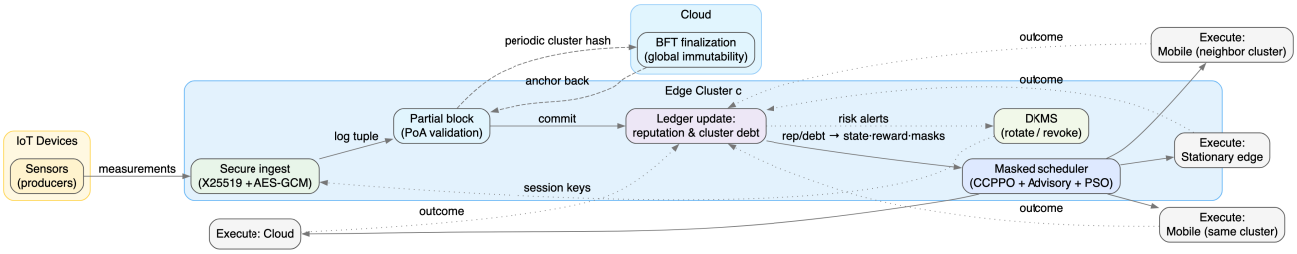


Fig. 1. End-to-end loop: secure ingest → PoA partial blocks → tamper-aware scheduling (masked CCPPO + advisory + PSO) → BFT finalization.

ogy.csv, and per-run *trace_epXX.csv* with KPIs (see artifact index in Section IV).

Algorithm 3 Deterministic Labeling & Deadline Assignment

Require: raw row (site, t, $\mathbf{c}$); thresholds $\boldsymbol{\tau}$; gradient window Δt; PRNG seed s

Ensure: task record $\{t_i, b_i, k_i, m_i, p_i, d_i, \text{loc}_i, \text{tier}_i, \text{prio}_i\}$

Feature extraction

$z \leftarrow \text{RobustZ}(\mathbf{c})$ ▷per-pollutant robust z-scores

grad $\leftarrow \text{RateFlags}(\mathbf{c}, \Delta t)$ ▷rate-of-change / trend flags

Tiering and traffic class

$\text{tier}_i \leftarrow \text{TierLookup}(z, \text{grad}; \boldsymbol{\tau})$ ▷$\in$ {Immediate, Routine, Delayed}

$(p_i, \mathcal{D}_i) \leftarrow \text{ClassAndDeadlineBand}(\text{tier}_i)$ ▷$p_i \in$ {URLLC, eMBB, mMTC}; $\mathcal{D}_i = [d_{\min}, d_{\max}]$

Resource footprints

$(b_i, k_i, m_i) \leftarrow \text{FootprintModel}(z, \text{grad})$ ▷payload (bytes), CPU cycles, memory

Deadline and priority

$d_i \leftarrow \text{Uniform}(\mathcal{D}_i;\ s)$ ▷deterministic via fixed seed s

$\text{prio}_i \leftarrow \text{Priority}(z;\ \text{hysteresis})$ ▷exceedance magnitude with hysteresis

Emit

return $\{t_i{=}t,\ b_i,\ k_i,\ m_i,\ p_i,\ d_i,\ \text{loc}_i{=}\text{site},\ \text{tier}_i,\ \text{prio}_i\}$

8) Policy Table for Tiers: Thresholds per pollutant ($PM_{2.5}$, PM_{10}, NO_2, O_3, SO_2) and temporal gradients drive tiering. We encode it as a deterministic table applied before any learning to prevent label leakage.

9) Why These Choices: 1) WHO-aligned tiering yields interpretable deadlines [40]; 2) fixed shares across scales isolate *load* as the only varying factor for OBJ1–OBJ4; 3) deterministic labeling avoids train/eval leakage; 4) standardized observations stabilize CCPPO across regimes.

C. City-Scale Simulator and Topology

1) Footprint and Topology: We simulate a 100 km^2 smart city partitioned into $C = 10$ clusters via *weighted* K-means (density + exceedance weight). Each cluster c hosts stationary and mobile edge servers; the cloud is $c = 0$. Neighbor sets $\mathcal{N}(c)$ follow Delaunay adjacency on cluster centroids. As shown in Fig. 2, clusters (red $\times$) are overlaid on a risk heatmap with device and server placements.

2) Network and Link Model: Effective per-link goodput $\bar{R}_{i,s}$ combines wireless and backhaul factors

$$\bar{R}_{i,s} = \zeta_{\text{slice}} \cdot \frac{W}{\Phi} \log_2\big(1 + \text{SNR}_{i,s}\big) \cdot \eta_{\text{MAC}} \cdot \eta_{\text{backhaul}} \tag{30}$$

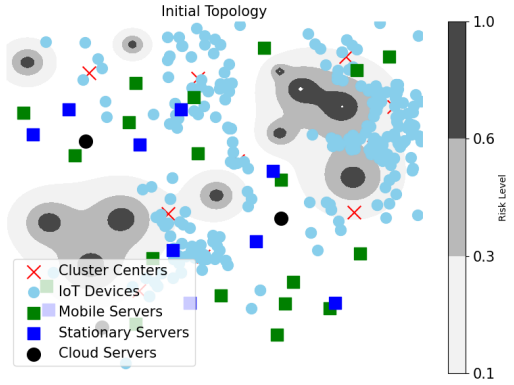


Fig. 2. Initial topology and risk heatmap for the 100 km^2 area. Red $\times$: cluster centers; light-blue dots: IoT devices; green squares: mobile servers; blue squares: stationary servers; black circles: cloud servers. Shaded grayscale indicates environmental risk/obstacles.

where W is per-slice bandwidth, Φ is frequency reuse, η_{MAC} captures contention, and η_{backhaul} captures cross-cluster/core load. Slices consume *time budgets* $\Gamma^{\text{radio}}_{c,k}$ (see Section III-C); mask construction forbids allocations that would exceed budgets within horizon H.

3) Queue and Processing Model: Each executor s uses a calibrated single-server surrogate

$$L^{\text{queue}}_{i,s} \approx \frac{\rho_s}{1-\rho_s} \cdot \frac{c_a^2 + c_s^2}{2} \cdot \mathbb{E}[S_s] \tag{31}$$

$$L^{\text{proc}}_{i,s} = \frac{k_i}{f_s} \tag{32}$$

with ρ_s from live load, service-time coefficient c_s estimated online, and arrival variability c_a from the task generator.

4) Mobility and Routing of Mobiles: Mobile servers follow conflict-based search (CBS)–D* Lite (Algorithm 4) on a grid overlaid with a risk/obstacle map [41]. Motion latency $L^{\text{move}}_{i,s}$ uses the current path length over v_s (up to 120 km/h). Collisions are prevented by CBS time-window constraints. As illustrated in Fig. 3, the planner redeploys mobiles from nearby clusters toward distress clusters while detouring around higher-risk cells.

5) Path-Planning Visualization and Analysis: Fig. 3 shows CBS–D* Lite paths used by *EdgePlus* to redeploy mobile servers from nearby clusters toward distress clusters over a risk/obstacle heatmap (darker = higher risk). The planner finds feasible, collision-free routes without deadlocks, detouring around elevated-risk cells when needed. Across the seven deployments shown, all mobiles reach their goals while avoid-

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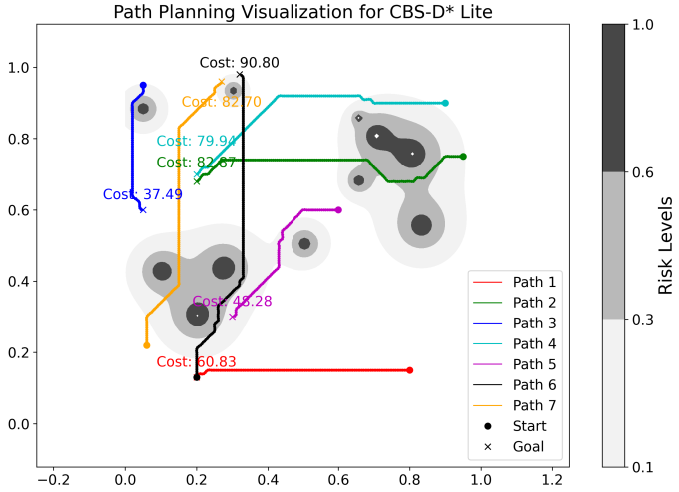


Fig. 3. CBS–D* Lite path plans for seven mobile servers (distinct colors) from nearby clusters to distress clusters over a risk heatmap (darker = higher risk). All paths are feasible and deadlock-free; annotated costs indicate path lengths.

Algorithm 4 Integrated Path Planning (CBS + D* Lite)

Require: Agents $\mathcal{A} = \{1, \ldots, M\}$ with start–goal pairs $\{(S_j, G_j)\}_{j=1}^{M}$; grid graph $G = (V, E)$; risk map ψ; weights $(w_d, w_{\text{wait}}, w_r)$
Ensure: Conflict-free paths $\{P_j\}_{j=1}^{M}$ minimizing $\sum_j C(P_j)$ per Eqs. (33)–(34)

Step 1: Root initialization (low-level plans).
Build root constraints $\Omega \leftarrow \varnothing$; for each j, compute $P_j \leftarrow$ D* Lite$(S_j, G_j; \Omega)$ using c^{Ω} (Eq. (37)).
Push root node $N_0 = \left(\Omega, \{P_j\}, f = \sum_j C(P_j)\right)$ into a min-heap OPEN keyed by f.
Step 2: CBS loop (branch on earliest conflict).
while OPEN not empty **do**
  $N \leftarrow$ pop-min(OPEN); $(\Omega, \{P_j\}) \leftarrow N$.
  Scan $\{P_j\}$ for earliest conflict (a, b, t).
  if no conflict **then**
    return $\{P_j\}$feasible and minimal under Eqs. (35)–(36)
  for $q \in \{a, b\}$ **do**branch: forbid one side at time t
    $\Omega' \leftarrow \Omega \cup${forbid agent $q$at t}time-window constraint
    $P'_q \leftarrow$ D* Lite$(S_q, G_q; \Omega')$ with edge cost c^{Ω}.
    if P'_q exists **then**
      $N' \leftarrow \left(\Omega', \{P_{j \neq q}\} \cup \{P'_q\}, f' = \sum_j C(P'_j)\right)$; push N' into OPEN.
return INFEASIBLE no joint plan satisfies constraints
Step 3: Path optimization (optional smoothing).
Fit cubic splines to each P_j for visualization/playback.
Step 4: MARL policy coupling.
Update costs/priorities in the scheduler π using $C(P_j)$ and L^{move}; keep feasibility masks unchanged.

ing high-risk regions. The annotated costs reflect the resulting distance tradeoffs.

*a) CBS–D** Lite:* Let $G = (V, E)$ be the grid over the risk/obstacle heatmap and $\mathcal{T} = \{0, \ldots, H\}$ discrete time. Agent $j \in \{1, \ldots, M\}$ follows a time-indexed path $P_j = (x_{j,0}, \ldots, x_{j,T_j})$ with $x_{j,t} \in V$, $x_{j,0} = S_j$, $x_{j,T_j} = G_j$, and $(x_{j,t}, x_{j,t+1}) \in E \cup \{\text{wait}\}$. Edge cost combines geometric length, optional waiting, and risk bias

$$c(u, v) = w_d \, \|p(v) - p(u)\|_2 + w_{\text{wait}} \, \mathbf{1}[v{=}u] + w_r \, \psi(v) \tag{33}$$

yielding per-agent path cost

$$C(P_j) = \sum_{t=0}^{T_j - 1} c\left(x_{j,t}, x_{j,t+1}\right). \tag{34}$$

Agent feasibility on the time-expanded graph enforces standard CBS conflict rules for all $a \neq b$ and all t

$$\text{(vertex)} \quad x_{a,t} \neq x_{b,t} \tag{35}$$

$$\text{(edge-swap)} \quad \left(x_{a,t}, x_{a,t+1}\right) \neq \left(x_{b,t+1}, x_{b,t}\right). \tag{36}$$

CBS searches a constraint tree over Ω (forbidden $\langle j, t, \cdot \rangle$ vertices/edges). At each node, low-level planning uses D* Lite with the *constrained* edge cost

$$c^{\Omega}(u, v) = \begin{cases} +\infty, & (u \to v) \text{ violates } \Omega \text{ or keep-outs} \\ c(u, v), & \text{otherwise} \end{cases} \tag{37}$$

and the node objective is the sum of individual costs

$$\min_{\{P_j\}} \sum_{j=1}^{M} C\left(P_j\right)$$

$$\text{s.t.} \quad (35)\text{–}(36), \quad \left(x_{j,t}, x_{j,t+1}\right) \in E \cup \{\text{wait}\} \quad \forall j, t. \tag{38}$$

The search procedure that enforces (35)–(36) is summarized in Algorithm 4. Branching adds a single disjunctive constraint to Ω for one conflicting agent (vertex or edge-swap), and replans via D* Lite; the best-cost consistent node is expanded next.

b) Feasibility Mask Construction: At each decision for task i, we build $M_{i,\cdot}$ in the routing order

$$\text{stationary in } c(i) \;\to\; \text{mobile in } c(i) \;\to\; \text{mobile in } \mathcal{N}(c(i)) \;\to\; \text{cloud}.$$

We then zero entries that violate any of: slice time budgets (radio/compute), per-server capacity/energy, deadline feasibility $t_i + L_{i,s} \leq d_i$ (hard for URLLC when safety gating is enabled), or low reputation for URLLC). The mask is the *only* mechanism that removes actions; advisory biases never override it. We use the five-episode protocol with fixed per-episode shares (see Table IV).

D. Compared Methods (Schedulers)

We benchmark *five* schedulers, all calling the same Environment/Evaluator API.

1) *ccppo_edgeplus (Full Method):* Coordinated/Constrained PPO (centralized critic) with *feasibility masks*; advisory (Nash/PID/Leader) enabled; constricted PSO enabled; *ledger feedback enabled*. Default EdgePlus unless ablated in OBJ5.
2) *ensemble_ppo [42]:* Bootstrap ensemble of PPO actors for nonstationary loads (inherits environment-level masks/ledger when toggled).
3) *fd3qn [43]:* Dueling double DQN with n-step targets (discrete actions; inherits masks/ledger when toggled).

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4) *ddpg [44]:* Off-policy actor–critic (continuous logits → discretized; inherits masks/ledger when toggled).
5) *sac [45]:* Entropy-regularized actor–critic with tuned temperature (inherits masks/ledger when toggled).

Used only in ablations (OBJ5): ccppo_lagr = Lagrangian CCPPO with *no masks*, advisory off, PSO off; identical to A6 (plain CCPPO) only when the ledger is also disabled.

E. Training and Inference Protocol

1) Budget and Splits: Hyperparameters are tuned once on 50 000 tasks and then *frozen* for 50 000/100 000/150 000 (OBJ3). For each scheduler, we fix seeds, episode horizons, and vectorized workers.

2) Centralized Critic Training on Global Context: A centralized critic V_ψ trains on global context; actors see only local observations at inference. We use PPO with GAE, clipping ϵ, entropy bonus $\lambda_{\mathcal{H}}$, and optional KL penalty λ_{KL}.

3) Stability: We apply gradient clipping $\|g\|_2 \leq g_{\max}$; observation standardization uses training statistics only; no early stopping. At evaluation, all policies are *frozen*.

4) PSO Outer Loop: When enabled for *ccppo_edgeplus*, constricted PSO runs either as 1) a pretuning phase that warm-starts (θ, ψ), or 2) short interleaves every K_{PSO} episodes; basins are accepted only if masked episodic fitness improves (with SLA-gated acceptance as in Section III-G). PSO is *disabled at evaluation*.

F. Metrics and KPIs (Tie to Objectives)

We report the KPIs in Table V. Their mapping to objectives is as follows.

1) *OBJ1 (Stress QoS):* Success, violations, $L_{p50/p95}$ under 150 000.
2) *OBJ2 (Light Load):* Success → 100%, seconds-level latency at 50 000.
3) *OBJ3 (Scaling):* Deltas across 50 000→ 100 000 and 100 000→ 150 000, and overall 50 000→ 150 000.
4) *OBJ4 (Locality):* LI, χ_{cloud}, and correlation with violations.
5) *OBJ5 (Ablations):* KPI deltas when toggling masks, advisory, PSO, and ledger feedback at 50 000 and 150 000.
6) *OBJ6 (Control):* PID stability envelope (Jury test), overshoot M_p, settling time T_s under ±20% gain drift.
7) *OBJ7 (Security):* Defense/detection for CIA/auth; DKMS rotate/revoke overhead; ledger consensus cost and throughput.

For central tendency we report means with 95% CI over episodes; dispersion via IQR and p5/p95 where relevant. Pairwise comparisons use Wilcoxon signed-rank; multiple testing is Benjamini–Hochberg corrected.

V. Results

A. OBJ1—Stress QoS (150k Tasks)

1) Definition: Stress QoS asks how well each scheduler preserves quality of service when the city is saturated with 150 000 tasks. Concretely, we care about meeting deadlines

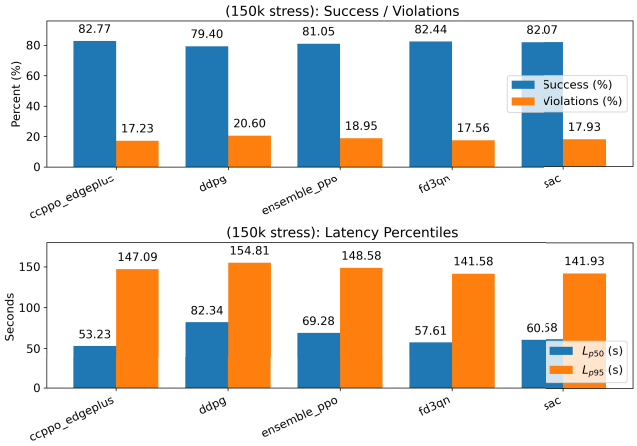


Fig. 4. OBJ1 (150k): reliability (success/violations) and latency percentiles across schedulers.

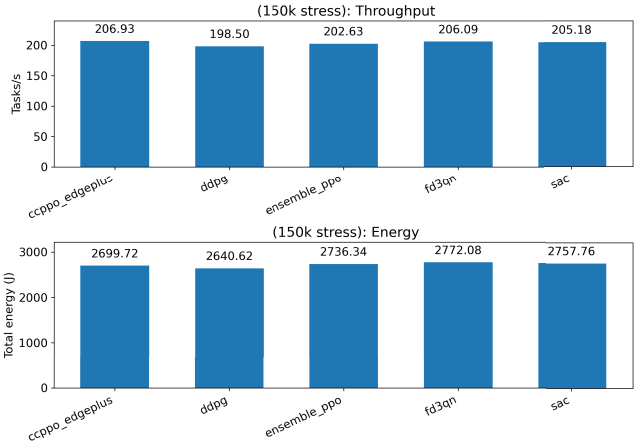


Fig. 5. OBJ1 (150k): throughput (tasks/s) and total energy across schedulers.

(reliability) while avoiding pathological tails and excessive resource burn.

2) Metrics: We report the reliability and efficiency indicators from Table V: on-time *success* σ and *violation* rate; latency at the median and tail (L_{p50}/L_{p95}); *throughput* (tasks/s); and total *energy*. Throughout, policies are frozen at inference (see Section IV-E); error bars (when shown) are 95% CIs over five episodes.

3) Experiment: All schedulers operate under identical topology, slice budgets, and seeds (see Section IV). Feasibility masks, advisory, PSO, and ledger feedback are enabled or disabled according to each method's definition (see Section IV-D); the PSO outer loop is *disabled* during evaluation to isolate policy behavior. We run five stress episodes of 150 000 tasks each and log the KPIs.

4) Results: Fig. 4 contrasts reliability and latency. *ccppo_edgeplus* achieves the highest on-time completion with the fewest misses, while keeping L_{p50} and L_{p95} competitive. The next best performers, *fd3qn* and *sac*, track closely on tails but concede a small, consistent gap in success. *ensemble_ppo* sits mid-pack, and *ddpg* trails with the heaviest tails under saturation.

Fig. 5 complements this view with throughput and energy. *ccppo_edgeplus* and *fd3qn* deliver the highest task throughput, evidence that constraint-faithful decisions do not come at the

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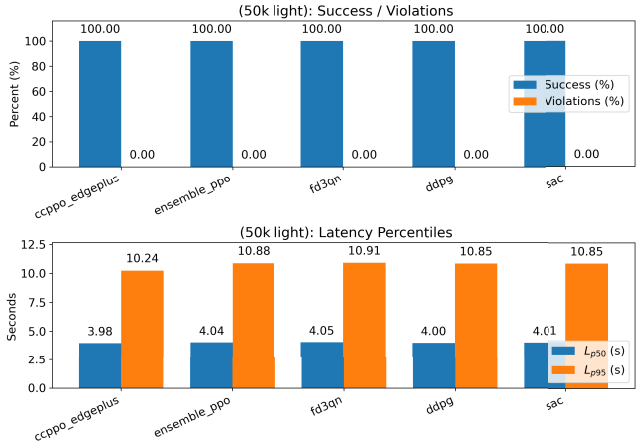


Fig. 6. OBJ2 (50k): success (near-100% across all schedulers) and latency percentiles.

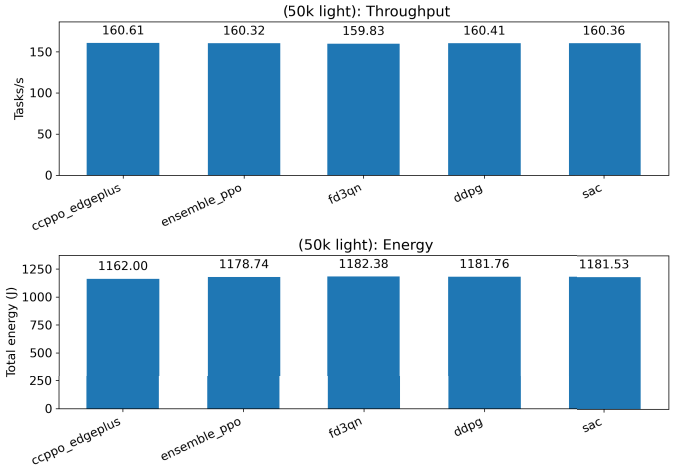


Fig. 7. OBJ2 (50k): throughput (tasks/s) and total energy.

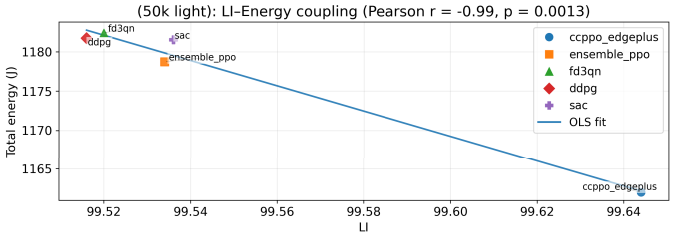


Fig. 8. OBJ2 (50k): LI–energy coupling; higher locality corresponds to lower energy.

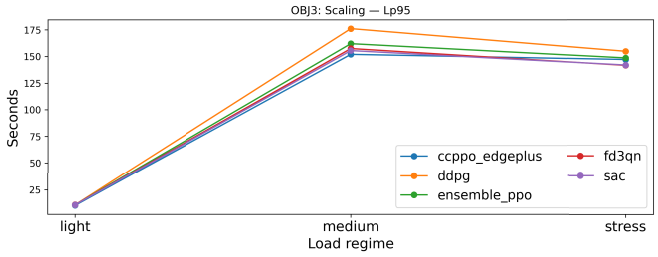


Fig. 9. OBJ3: tail latency L_{p95} across load regimes (50k, 100k, 150k). Lower is better.

cost of system utilization, while *ddpg* spends the least energy but does so alongside weaker reliability and larger tails. Taken together, these curves indicate that *ccppo_edgeplus* offers the best reliability–latency–throughput balance at city scale, with the strongest gains appearing exactly where saturation stressors are most acute (URLLC gating, slice adherence, and queue pressure).

5) Takeaway: Under peak load, *ccppo_edgeplus* leads on the primary objective, which is meeting deadlines, without incurring worse tails, and it sustains top-tier throughput. The pattern is consistent across seeds and aligns with the feasibility-first design of EdgePlus (masks + advisory), which prioritizes admissible actions before optimization.

B. OBJ2—Light-Load Sanity (50k Tasks)

1) Definition: Light-load sanity verifies that, in a normal situation where congestion is absent, the scheduler is able to deliver near-perfect reliability with low latency and without incurring overheads from masking or security. This is our correctness check before scaling tasks' load.

2) Metrics: We reuse the KPIs in Table V: *success* σ and *violations* for reliability; L_{p50}/L_{p95} for latency; *throughput* (tasks/s) and *energy*. We additionally examine the LI and its coupling to energy at low load.

3) Experiment: All methods are evaluated with 50 000 tasks, identical topology and seeds (see Section IV), and policies frozen at inference (see Section IV-E). The PSO outer loop remains disabled at evaluation. Figs. 6 and 7 summarize reliability/latency and throughput/energy; Fig. 8 visualizes LI–energy coupling.

4) Results: As shown in Fig. 6, every scheduler reaches $\sigma \approx 100\%$ with seconds-scale latency (median ~4 s and $L_{p95} \sim 11$ s), indicating no artificial slowdowns from feasibility masks or the security path at this load. Throughput and energy are comparable across methods (see Fig. 7); minor differences reflect implementation details rather than structural tradeoffs. The LI–energy scatter in Fig. 8 reveals a strong negative correlation (higher locality $\Rightarrow$ lower energy), consistent with intuition that keeping work in-cluster avoids backhaul and motion overheads.

5) Takeaway: Under light task load, all schedulers behaved as expected: deadlines were met, tails were short, and overheads from feasibility and security remained negligible. This established a clean baseline for the scaling and stress analyses that followed.

C. OBJ3—Scaling With Load (50k → 100k → 150k)

1) Definition: Scaling with load asks how gracefully each scheduler degrades as offered work rises. We care about whether reliability falls slowly, tails remain bounded, and placement adapts without over-reliance on the cloud.

2) Metrics: We track the KPIs in Table V—*success* σ, *violations*, and latency percentiles (L_{p50}/L_{p95}), and summarize sensitivity using stepwise deltas across three transitions: LM (50k→ 100k), MS (100k→ 150k), and LS (50k→ 150k). For interpretability, we discuss a *QoS elasticity* notion: smaller increases in $(1 - \sigma)$ for a given load step indicate better elasticity.

3) Experiment: All schedulers are evaluated at 50 000, 100 000, and 150 000 with identical seeds and topology (see

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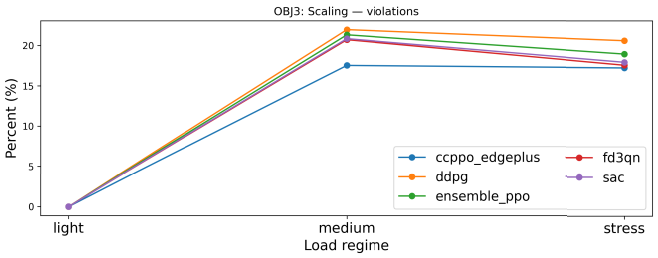


Fig. 10. OBJ3: deadline violations across different load regimes. The lower the better.

Section IV); policies are frozen at inference (see Section IV-E). The PSO outer loop is disabled during evaluation for everyone. Figs. 9 and 10 plot latency tails and violations across the three loads.

4) Results: The sharpest change occurs at the LM step. Moving from 50k to 100k increases L_{p95} for all methods (see Fig. 9); at 100k, ddpg exhibits the heaviest tails (∼175 s), while ccppo_edgeplus, fd3qn, sac, and ensemble_ppo cluster lower (∼150–162 s). This same LM transition also drives the largest jump in deadline violations (see Fig. 10). Interestingly, the MS step shows mild tail *compression* for every method (slightly lower L_{p95} at 150k than at 100k), consistent with tighter feasibility masks and more conservative admission under pressure; violations relax modestly in tandem.

The ordering is stable at higher loads. ccppo_edgeplus sustains the lowest violations (about 17.5% at 100k and 17.2% at 150k) with competitive tails (∼151 s at 100k; ∼145 s at 150k). fd3qn and sac closely track those outcomes. ensemble_ppo sits mid-pack, while ddpg remains dominated on both tails and violations across 100k and 150k. LI dips around 100k and partially recovers at 150k, an effect we unpack next in OBJ4 (see Section V-D).

5) Takeaway: Capacity pressure bites hardest in the first jump (50k→ 100k). Under sustained load, feasibility-first scheduling compresses tails and stabilizes violations. By this elasticity lens, ccppo_edgeplus degrades most gracefully, with fd3qn/sac close behind and ddpg consistently lagging.

D. OBJ4—Locality, Reliability, and Offload

1) Definition: *Locality* captures how often tasks finish where they originate. We quantify it with the LI: the fraction of tasks executed in their home cluster $c(i)$. The question is whether higher LI buys better deadline reliability and less dependence on the cloud.

2) Metrics: We use the KPIs in Table V: LI, the cloud-offload ratio χ_{cloud}, and the deadline-violation rate. We analyze associations with Pearson/Spearman correlations and a robust Theil–Sen slope to express "pp change in χ_{cloud} or violations per + 1 pp LI."

3) Experiment: For each method × load point (light/medium/stress), we compute LI, χ_{cloud}, and violations over the same five-episode protocol (see Section IV). We report aggregate correlations across all points and then stratify by load. Fig. 11 illustrates the LI↔ χ_{cloud} relationship across loads (Theil–Sen fit in-line). Fig. 12 focuses on LI versus violations at 150 000.

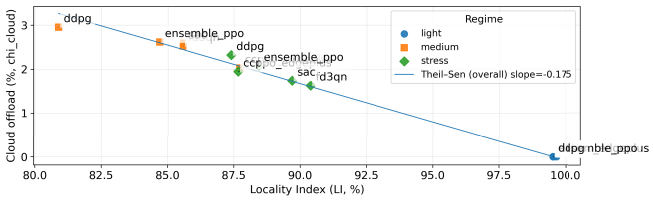


Fig. 11. OBJ4: LI versus cloud offload across loads. Each point is a method × load; robust Theil–Sen line shown. Higher LI consistently reduces χ_{cloud}.

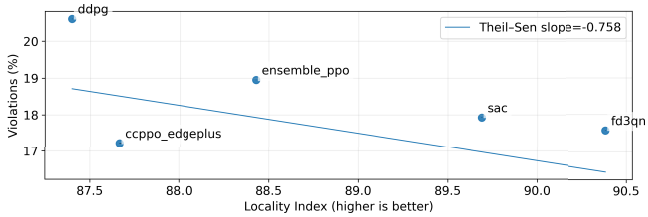


Fig. 12. OBJ4: LI versus deadline violations at 150 000. Negative trend visible but not statistically significant with $n = 5$.

4) Results: Across all methods and loads, LI is *tightly* anti-correlated with cloud use (Pearson $r = -0.995$, $p < 10^{-12}$; Spearman $\rho = -0.94$, $p \approx 10^{-6}$). The Theil–Sen estimate indicates each + 1 pp LI reduces χ_{cloud} by ≈ 0.175 pp (95% CI $[-0.18, -0.13]$); see Fig. 11. LI also anticorrelates with deadline violations overall (Pearson $r = -0.77$, $p = 8.4 \times 10^{-4}$; Spearman $\rho = -0.77$, $p = 9.4 \times 10^{-4}$) with a slope of ≈ −0.76 pp violations per + 1 pp LI (CI $[-1.05, -0.54]$).

Stratified by load, the picture remains consistent: at *stress* (150k), LI versus cloud is again extremely strong (Pearson $r = -0.997$, $p = 0.0017$; Theil–Sen ≈ −0.231 pp cloud per +1 pp LI), while LI versus violations shows a negative but sample-limited trend (Pearson $r = -0.52$, $p = 0.37$; Fig. 12). At *medium* load (100k), LI retains strong inverse links to both cloud (Pearson $r = -0.93$, $p = 0.021$; Spearman $\rho = -0.90$, $p = 0.037$) and violations (Spearman $\rho = -0.90$, $p = 0.037$). At *light* load (50k), cloud usage is near zero for most methods, so correlations are uninformative.

5) Takeaway: Keeping work close to where it arrives materially reduces cloud dependence and, in aggregate, tracks fewer deadline misses. This validates EdgePlus's feasibility-first routing order and locality-aware advisory bias: by preferring in-cluster execution when feasible, we cut backhaul reliance and protect QoS without sacrificing tails (see Figs. 11 and 12).

E. OBJ5—Ablations: Masks, Advisory, PSO, and Ledger

1) Definition: This objective disentangles where EdgePlus's gains come from. We remove one component at a time, feasibility *masks*, the *advisory* layer (Nash/PID/Leader), the *PSO* outer loop, and *ledger* feedback, and observe how reliability, latency, locality, and cloud use respond.

2) Metrics: We track the same KPIs as in Table V: success σ, violation rate, L_{p50}/L_{p95}, throughput, LI, and χ_{cloud}. When comparing against the full system, we emphasize *deltas* in violations (pp), L_{p95} (s), LI (pp), and cloud share (pp).

3) Experiment: Starting from full ccppo_edgeplus (A1), we evaluate five additional configurations (A2–A6) that

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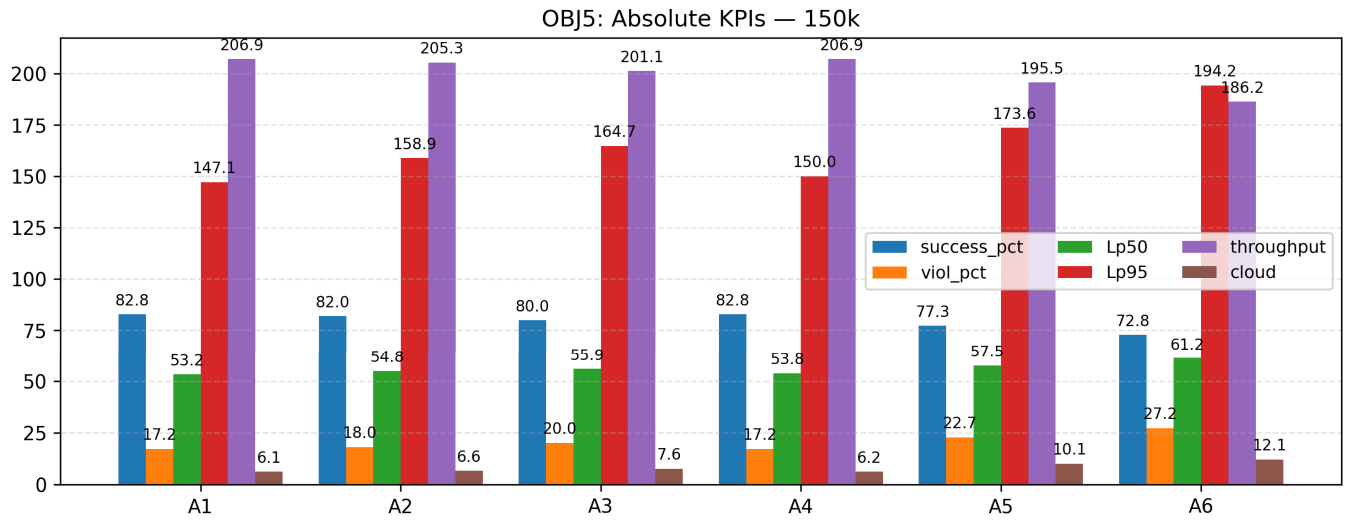


Fig. 13. OBJ5 at 150 000: absolute KPIs for A1–A6. Higher is better for success/throughput; lower is better for violations, L_{p50}/L_{p95}, and cloud share.

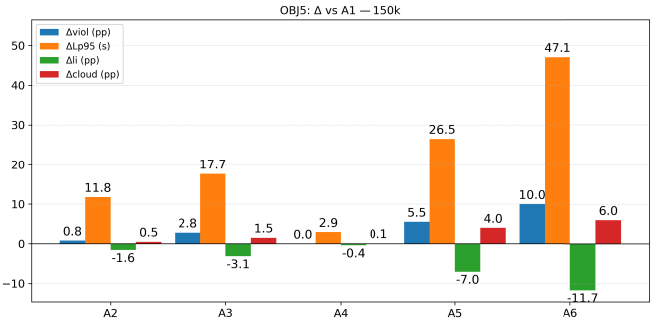


Fig. 14. OBJ5 at 150 000: deltas versus A1 (full EdgePlus). Bars show Δviol (pp), ΔL_{p95} (s), ΔLI (pp), and Δcloud (pp).

TABLE VI
OBJ5 ABLATION CONFIGURATIONS. A1 IS THE FULL EDGEPLUS STACK

Config	Masks	Advisory	PSO	Ledger
A1 (full EdgePlus)	✓	✓	✓	✓
A2 (no PSO)	✓	✓	–	✓
A3 (no advisory)	✓	–	✓	✓
A4 (no ledger)	✓	✓	✓	–
A5 (no masks)	–	✓	✓	✓
A6 (plain CCPPO)	–	–	–	–

selectively disable components (see Table VI). We run the standard five-episode protocol (see Section IV) at both 50 000 and 150 000. Figs. 13 and 14 report absolute KPIs and deltas at 150 000, respectively; light-load results mirror OBJ2 (near-100% success across the board) and are omitted for brevity.

4) Results: At 150 000, the full system (A1) achieves 82.8% success with 17.2% violations, L_{p50} = 53.2 s, L_{p95} = 147.1 s, throughput 206.9 tasks/s, and cloud share 6.1% (see Fig. 13). Removing PSO (A2) nudges the tail and cloud upward, L_{p95} +11.8 to 158.9 s, violations +0.8 pp to 18.0%, cloud +0.5 pp to 6.6%, while leaving success/throughput largely intact. Turning off the advisory layer (A3) amplifies those degradations: success falls to 80.0% (+ 2.8 pp violations), L_{p95} rises to 164.7 s (+17.7 s), and cloud grows to 7.6% (+1.5 pp). Disabling ledger feedback (A4) keeps success and throughput on par with A1 but modestly worsens placement/tails (L_{p95}150.0 s, + 2.9 s; cloud 6.2%, + 0.1 pp), consistent with softer reputation/debt signals.

The largest regressions appear when feasibility masks are removed. Without masks (A5), success drops to 77.3 % (+5.5 pp violations), tails stretch to 173.6 s (+ 26.5 s), cloud nearly doubles to 10.1% (+ 4.0 pp), and LI sinks by 7.0 pp. Removing everything (A6, plain CCPPO) is worst overall: success 72.8% (+ 10.0 pp), L_{p95} = 194.2 s (+ 47.1 s), throughput 186.2 tasks/s, cloud 12.1% (+ 6.0 pp), and LI −11.7 pp; see deltas in Fig. 14.

5) Takeaway: Masks are foundational: enforcing feasibility at sampling time dominates reliability and tail behavior. The advisory layer and PSO each add measurable headroom, advisory by biasing feasible choices toward fair, stable operation; PSO by escaping local optima during training. Ledger feedback is a lighter boost that helps placement and tails without shifting raw throughput. Together, these layers produce the reliability/locality profile of A1 (see Figs. 13 and 14).

F. OBJ6—Control Robustness (PID Under ±20% Gain Drift)

1) Definition: This objective asks whether EdgePlus's utilization controller remains stable and responsive when its PID gains drift (for instance, due to temperature or aging). We model the plant as a first-order plus dead time (FOPDT) process and close the loop with a discrete PID, then sweep gain perturbations.

2) Metrics: We deem a controller *feasible* if the unit-step closed loop settles within ±2% in $T_s \le 10$ s with overshoot $M_p \le 15\%$. We also report the energy of the control effort (arbitrary units) to capture aggressiveness.

3) Experiment: We discretize an FOPDT plant

$$G(s) = \frac{K}{\tau s + 1} e^{-Ls} \tag{39}$$

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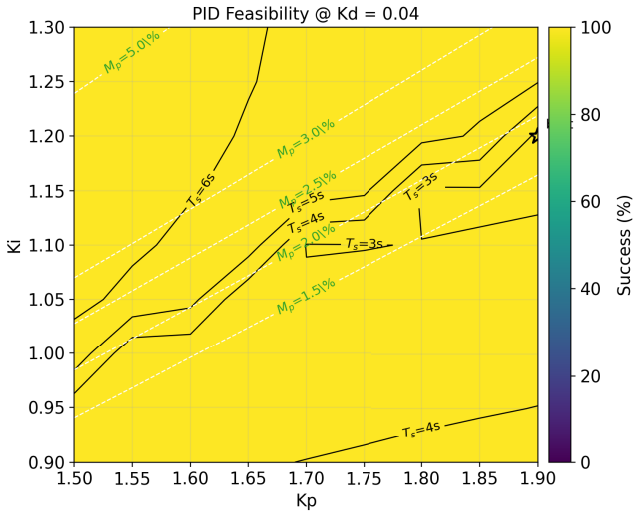


Fig. 15. OBJ6: feasibility under ±20% PID gain drift (k_d = 0.04). Color: pass rate (0–100%). Solid black curves: settling-time isolines (T_s). Dashed white curves (green labels): overshoot isolines (M_p).

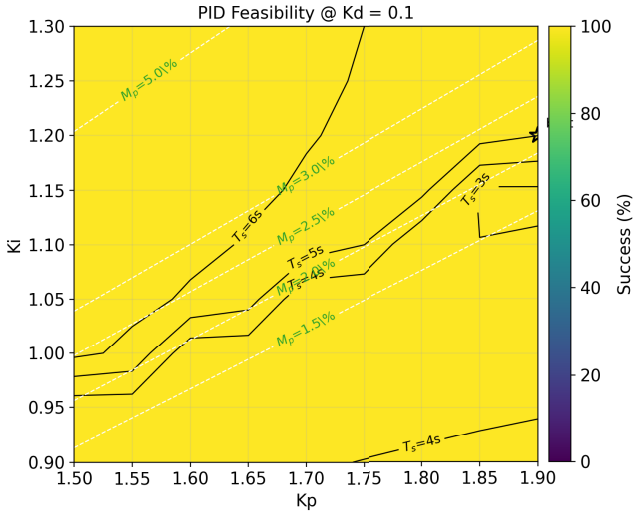


Fig. 16. OBJ6: feasibility under ±20% PID gain drift (k_d = 0.10).

TABLE VII

OBJ6 SUMMARY (90 s HORIZON; ±2% BAND; ANTI-WINDUP= 100; $u_{\max}$ = 100)

	k_p	k_i	k_d
Best (min T_s)	1.90	1.20	0.06
Robust pocket	[1.85, 1.90]	[1.15, 1.20]	[0.06, 0.12]

Best metrics: $T_s = 2.74$ s, $M_p = 1.93\%$, Energy = 92.88.

and form $P(z) = 1 + C(z)G(z)$ with a standard discrete PID $C(z)$. We sweep the ±20% hyper-rectangle around nominal gains

$$\mathcal{K} = \left\{ \left(K_p', K_i', K_d'\right) : \; K_\bullet' = (1 + \delta_\bullet) K_\bullet, \quad |\delta_\bullet| \le 0.2 \right\} \tag{40}$$

screen candidates with Jury's stability test, then simulate a 90 s unit-step with $u_{\max}$ = 100 and anti-windup clamp = 100. Figs. 15–17 visualize feasibility for three fixed k_d slices; Table VII summarizes the best setting and a robust pocket.

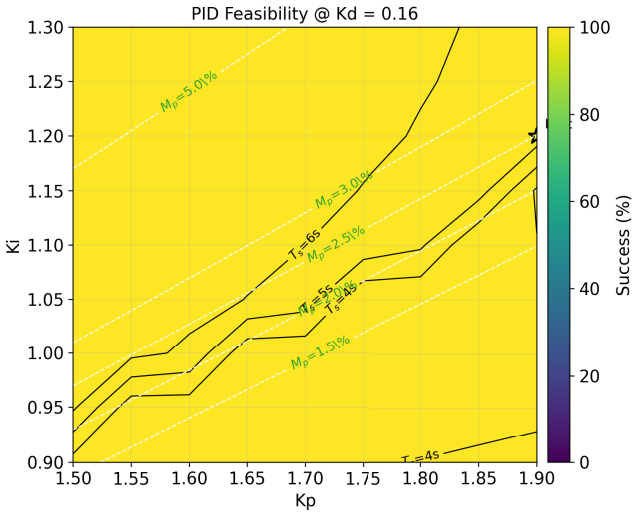


Fig. 17. OBJ6: feasibility under ±20% PID gain drift (k_d = 0.16).

4) Results: A tight, conservative center sweep produces 0% feasible points (settling is capped by the horizon). Allowing more authority and widening the search yields a 53.6% pass rate (2443/4560 feasible triplets). Local refinement identifies a pocket that is both fast and robust: the best triplet $(k_p, k_i, k_d) = (1.90, 1.20, 0.06)$ achieves $T_s = 2.74$ s, $M_p \approx 1.93\%$, energy ≈92.88, and neighboring settings $k_p \in [1.85, 1.90]$, $k_i \in [1.15, 1.20]$, $k_d \in [0.06, 0.12]$ maintain $T_s \approx 2.76$–2.88 s and $M_p \approx 1.7\%$–2.0% (see Table VII).

Interpreting Figs. 15 –17. Each figure fixes k_d and scans a grid of (k_p, k_i). Color indicates the *fraction of ±20% drift realizations that pass* the $\{T_s, M_p\}$ criteria (yellow ≈ 100% pass). Solid *black* isolines mark equal settling time ($T_s \in \{3, 4, 5, 6\}$ s from bottom to top). Dashed *white* isolines (green labels) mark equal overshoot ($M_p \in \{1.5\%, 2.0\%, 2.5\%, 3.0\%, 5.0\%\}$ increasing up/left).

1) *Derivative Action Widens the Safe Band:* Comparing Fig. 15 (k_d = 0.04) to Fig. 16 (k_d = 0.10), the T_s = 3–4 s contours shift *down-right*: the same settling can be achieved with slightly smaller k_p or k_i, and the yellow (high-pass) region expands. Fig. 17 shows the trend continuing but requiring a touch more k_i to keep $T_s \le 4$ s.
2) *Too Little Integral Slows Settling:* At low k_i (bottom rows), the T_s isolines rise quickly, settling degrades before overshoot becomes a concern.
3) *Too Much Proportional Risks Overshoot:* At high k_p (right edge), overshoot isolines drift upward: marginal M_p increases constrain how far right one can go without crossing the 15% cap.
4) *Robust Pocket:* Across figures, the broadest yellow band centers around $k_p \approx 1.85$–1.90 and $k_i \approx 1.15$–1.20. This is where our chosen $(1.90, 1.20, 0.06)$ sits; increasing k_d from 0.04 to 0.10 preserves this pocket's width, while k_d = 0.16 nudges it slightly upward in k_i.

5) Takeaway: EdgePlus can be tuned to a *locally robust* operating region where ±20% gain drift preserves tight settling and low overshoot. Practically, this means the advisory PID

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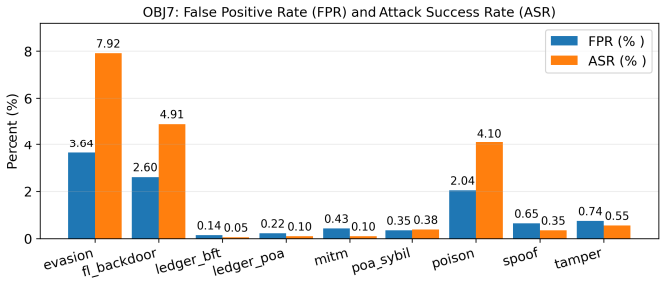


Fig. 18. OBJ7: FPR and ASR by attack type at 150 000. Lower is better.

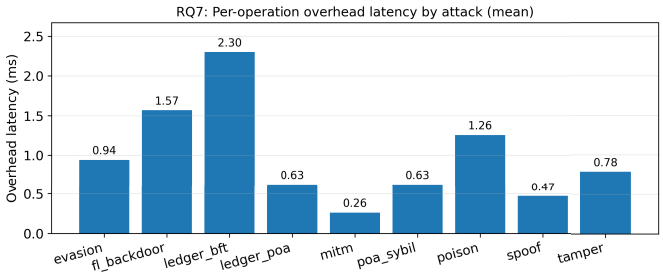


Fig. 19. OBJ7: peroperation overhead latency (median, ms) by attack type at 150 000. Lower is better.

TABLE VIII

BASELINE THREATS: DEFENSE/DETECTION AND PEROPERATION OVERHEAD (MEDIAN; IQR IN PARENTHESES). WHERE ZERO MISSES OCCUR, 95% UPPER BOUNDS USE THE RULE OF THREE

Threat	Defense (%)	Detection (%)	Overhead (ms)
Eavesdropping (MITM)	99.2	99.0	0.18 (0.11–0.27)
Credential spoofing	98.9	99.3	1.12 (0.92–1.46)
Data tampering	99.1	99.4	0.42 (0.31–0.58)

can regulate cluster utilization without inducing oscillations, even as component characteristics vary.

G. OBJ7—Security Efficiency and Resilience

1) Definition: We examine whether EdgePlus can secure data and control paths end-to-end while preserving tight latency budgets. The focus is on confidentiality, integrity, authentication (CIA), ledger tamper-evidence, and key lifecycle (DKMS) under adversarial pressure.

2) Metrics: We report *defense* and *detection* rates (%), false-positive rate (FPR, %), attack success rate (ASR, %), per-operation *overhead* (ms), the security efficiency index (SEI = Defense%/median ms), consensus amplification factor (CAF), and *security tax* (added latency, ms). When no misses occur, 95% upper bounds use the rule of three.

3) Experiment: All links derive session keys via X25519 and protect payloads with AES–GCM; control-plane messages and blocks use Ed25519. The edge ledger validates *partial blocks* via PoA; the cloud periodically finalizes via BFT. We sweep adversary compute share $\lambda_{\text{adv}} \in \{0.05, 0.10, 0.25\}$ and vary validator counts (10/20/30) using sub-quorums for $C > 10$. Baseline CIA/auth threats (MITM eavesdropping, credential spoofing, data tampering) and advanced threats (evasion, FL backdoor, poisoning, PoA/BFT ledger tampering, and authority/Sybil) are injected. Results are summarized in Table VIII (baselines), Table IX (advanced), and visualized in Fig. 18 (FPR/ASR) and Fig. 19 (overhead).

4) Results: Baseline CIA/auth protections meet targets: *Defense* is 98%–99% with sub-ms to low-ms overheads (median 0.18–1.12 ms; Table VIII). Advanced, model-centric attacks are harder, evasion and FL backdoors show the highest residual ASR, but remain bounded with moderate overheads; poisoning sits in between (see Table IX, left columns; see also Fig. 18). Ledger attacks are efficiently contained: edge PoA achieves 99.8% defense with 0.39 ms median validation and the highest SEI, while cloud BFT provides accountable finality at higher cost (6.9 ms); corresponding CAF and Security Tax are reported in Table IX. Across threats, per-operation overheads remain within budget (see Fig. 19), and zero false-accept events on ledger paths translate to tight upper bounds via the rule-of-three.

5) Takeaway: EdgePlus delivers *tamper-aware* scheduling with low security tax: baseline CIA/auth is near-perfect at sub-ms to low-ms cost; PoA keeps edge validation sub-ms (0.39 ms median), and BFT finalizes globally without jeopardizing SLA windows. The remaining surface is predominantly model-level (evasion/backdoors), which we mitigate with abstention and robust aggregation while preserving real-time operation.

VI. DISCUSSION

A. What the Evidence Says

Across loads, ablations, and security trials, three themes recur. *First*, *feasibility-first* design is decisive. Removing masks (A5/A6) consistently degrades reliability, elongates tails, and increases cloud dependence (see Figs. 13 and 14). By contrast, masked CCPPO preserves routing order and slice budgets by construction, which is why ccppo_edgeplus maintains the lowest violation rates at 100k/150k while remaining competitive on L_{p95} (see Fig. 4). *Second*, lightweight coordination helps but is secondary: the advisory layer and constricted PSO each shave tail latency and cloud share (A2/A3 versus A1), yet their gains stack *on top of* masking rather than replacing it. *Third*, *locality matters*. The LI strongly anti-correlates with cloud offload and, in aggregate, tracks fewer deadline misses (see Fig. 11); when load peaks (150k), stricter masking/admission compresses tails and restores some locality (see Figs. 9 and 10).

On the control side, a simple PID with Ziegler–Nichols initialization and Jury certification yields a robust pocket under ±20% gain drift (see Figs. 15–17), giving sub-3 s settling without overshoot inflation, adequate for the episode dynamics we target. On the security path, the two-tier ledger meets a pragmatic goal: *sub-ms* PoA at the edge (median ≈ 0.39 ms) with periodic BFT finalization in the cloud (median ≈ 6.9 ms), keeping the per-operation security tax within budget while achieving 98%–99% mitigation on baseline CIA/auth and near-perfect containment for ledger tampering (see Tables VIII and IX and Fig. 19).

B. Design Lessons for Practitioners

Based on the experimental results across OBJ1–OBJ7, we distill the following practical design lessons for practitioners:

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TABLE IX
ADVANCED (NONPHYSICAL-LAYER) THREATS: EPISODE-AVERAGED DEFENSE/DETECTION, OVERHEAD (MEDIAN MS), SEI, CAF, AND SECURITY TAX (MS)

Threat	Defense (%)	Detection (%)	Overhead (ms)	SEI	CAF	Sec. Tax (ms)	Notes
Data poisoning (training)	80.7	82.1	12.4	6.5	–	–	Robust aggregation + ledger evidence; $\lambda_{\text{adv}} \in \{0.05, 0.10, 0.25\}$
Adversarial evasion (inference)	76.4	78.9	7.9	9.7	–	–	Score smoothing and abstention; no PHY attacks modeled
FL backdoor (federated)	73.2	75.6	10.6	6.9	–	–	Clipping, median-of-means, DKMS hygiene
Ledger tampering (edge PoA)	99.8	99.8	0.39	255.6	1.62	0.63	Sub-ms via sub-quorums to $C \leq 30$; zero false accepts
Ledger tampering (cloud BFT)	100.0	100.0	6.9	14.5	2.94	10.3	Periodic finalization; successful view-changes in all runs
Authority/Sybil against PoA	97.1	98.6	0.74	131.2	–	–	Enrollment hardening; quorum thresholding; fork tags

The results suggest a few simple rules of thumb.

1) Enforce Feasibility at Sampling Time: Use masks to encode routing order, slice quotas, resource and deadline checks; let learning optimize *within* the feasible set. This single choice dominated all other levers (see Fig. 14).
2) Bias, Do Not Override: Advisory signals (Nash/PID/Leader) should only reorder feasible actions. Gentle logits bias ($\lambda_N, \lambda_P, \lambda_L$) was enough to improve tails and locality without destabilizing training (A3 versus A1).
3) Treat PSO as a Safety-Gated "Escape Hatch": Interleave short, constricted PSO bursts with acceptance checks tied to SLA metrics (success, cloud share, stationary execution). This helps traverse poor basins (A2 versus A1) without regressing reliability.
4) Make Locality a First-Class KPI: Track LI and penalize avoidable cloud offload in the reward; higher LI coincides with better reliability and cost control (see Fig. 11).
5) SKeep the Ledger Lightweight at the Edge: Small validator sets or partitioned sub-quorums sustain *sub-ms* PoA; defer heavyweight finality to the cloud, amortized over time (see Table IX).
6) PStabilize Utilization With Certified PID: Pick gains from the robust pocket (see Figs. 15–17); if drift or thermal throttling emerges, back off K_p or raise the utilization setpoint and recertify via Jury.

C. Cost and Scalability

The complexity summary (see Table II) aligns with observed runtime: training cost scales linearly in active agents and constraints (masked CCPPO), while PSO appears only sporadically. Security overhead is dominated by BFT finalization, which is off the critical path; PoA validation remains in the sub-ms envelope (see Fig. 19). Practically, this means the end-to-end stack can be provisioned by sizing: 1) actor inference to decision-rate needs, 2) a modest PSO budget, and 3) a small, trusted edge authority set.

D. Limitations and Threats to Validity

Our study uses a city-scale simulator and a deadline-augmented air-quality workload; while representative of bursty sensing, it does not exhaust all 6G verticals. Queueing is modeled via calibrated surrogates (31); real deployments may exhibit heavier tails. Mobility relies on CBS–D* Lite; richer traffic and obstacle dynamics may further couple motion and radio conditions. Security experiments exclude physical-layer availability attacks by design; radio-aware defenses (e.g., RIS/jamming mitigation) are outside scope. Finally, the five-episode protocol keeps variance visible but limits statistical power for per-load stratifications (e.g., LI versus violations at 150k in Fig. 12).

E. Where This Transfers and What's Next

EdgePlus' invariants, masking for feasibility, advisory biasing, bilevel nonlocal exploration, and a ledger that feeds back reputation, are not tied to air quality. We expect similar behavior for vision or telemetry workloads that exhibit deadline tiers and locality pressure (e.g., traffic analytics, industrial inspection). Two immediate extensions are attractive: 1) integrating certified model robustness or selective abstention to further depress evasion/backdoor ASR (Table IX), and 2) moving from chance-constrained URLLC enforcement to formal CMDP safety with online certificates. Longer term, a hardware-in-the-loop testbed would exercise the cryptographic path and DKMS under real key-rotation storms, while multi-tenant slicing (with explicit interslice bargaining) could refine fairness beyond our current Jain metric.

F. Takeaway

In short, the simplest choices carried the day: enforce feasibility, keep coordination light, and make security *fast* where decisions are made. Everything else, PSO for escape, PID for smoothing, a ledger for accountability, then comes together into a scheduler that stays reliable at medium/stress

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loads (see Figs. 4 and 10) without paying a prohibitive security tax (see Fig. 19).

VII. Conclusion

This article introduced *EdgePlus*, a feasibility-first framework for deadline-sensitive 6G edge scheduling that unifies masked CCPPO, a lightweight advisory layer (Nash/PID/Leader), a safety-gated constricted PSO outer loop, and a two-tier blockchain-lite ledger. The design principle is simple: *make the feasible set explicit at sampling time*, then let learning and light coordination optimize *within* it.

Across a city-scale simulator and three load regimes, EdgePlus consistently delivered strong quality of service. Under light load (50 000), all methods achieved 100% deadline success with seconds-level latencies, confirming that masking and security do not introduce artificial delays. At stress load (150 000), EdgePlus attained the highest on-time completion (82.77%), highest throughput (206.93), lowest violation rate (17.23%), and lowest 15 percentile latency (L_{p50} of 53.23 s), while remaining competitive on tail latency L_{p95} (marginally trailing *fd3gn* and *sac*). Also, it preserved the lowest violation rates among peers and compressed tail latency via stricter feasibility/admission control. Locality emerged as a first-class operational KPI: higher LI coincided with lower cloud offload and, in aggregate, fewer deadline misses.

Ablations clarified where performance comes from. Removing masks dominated all regressions, raising violations by 5.5–10.0 pp, inflating L_{p95} by 26–47 s, and increasing cloud share by 4–6 pp, while the advisory layer and PSO provided additive but smaller gains. On the control path, a PID tuned via Ziegler–Nichols and certified by Jury's test sustained sub-3 s settling with $\pm 20\%$ gain drift. On the security path, the ledger met practical budgets: *sub-ms* PoA validation at the edge and periodic cloud BFT finalization, achieving 98%–99% mitigation for baseline CIA/auth attacks and near-perfect containment for ledger tampering, with low per-operation overheads.

A. Takeaways

1) Encoding constraints as masks and guard SLA-critical shares.
2) Use advisory signals to bias, not override, feasible choices.
3) Treat nonlocal search as an SLA-gated escape hatch rather than a primary optimizer.
4) Keep security *fast* where decisions are made and push stronger finality off the critical path. Together, these choices yield a scheduler that remains reliable at scale without paying a prohibitive security tax.

B. Broader Outlook

Although instantiated on air-quality workloads, EdgePlus's invariants, feasibility masking, advisory biasing, bilevel exploration, and reputation-aware feedback can be applied to other 6G verticals (such as traffic analytics, industrial inspection) that share deadline tiers and locality pressure. We release artifacts (datasets, execution logs, and supplementary figures) to encourage comparative studies and to support future extensions.

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